

Work and Kinetic energy

- ❑ The Scalar Product of Two Vectors
- ❑ Work
- ❑ Kinetic Energy
- ❑ Work and Kinetic Energy
- ❑ Elastic potential energy
- ❑ Gravitational potential energy



The Scalar Product

- Work is conveniently characterized using the *scalar product*, a way of combining two vectors to produce a scalar that depends on the vectors' magnitudes and the angle between them.
- The scalar product of two vectors $\vec{A}$ and $\vec{B}$ is defined as

$$\vec{A} \cdot \vec{B} = AB \cos \theta$$

where A and B are the magnitudes of the vectors and θ is the angle between them.

With vectors in component form, $\vec{A} = A_x \hat{i} + A_y \hat{j} + A_z \hat{k}$ and $\vec{B} = B_x \hat{i} + B_y \hat{j} + B_z \hat{k}$, the scalar product can be written

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z$$

- Work is the scalar product of force with displacement:

$$W = \vec{F} \cdot \Delta \vec{r}$$

Discuss the sign, magnitude of W as a function of angle

Textbook 6.1 part II

July 30, 2021



Scalar (Dot) Product of 2 Vectors

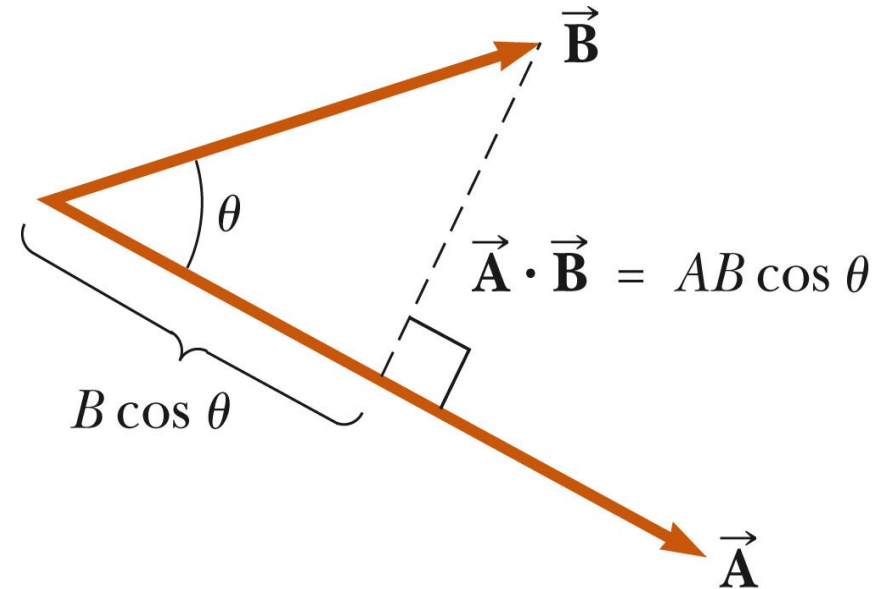
- The scalar product of two vectors is written as $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}}$

- It is also called the dot product

- $\vec{\mathbf{A}} \cdot \vec{\mathbf{B}} \equiv A B \cos \theta$

- θ is the angle *between* A and B

- Applied to work, this means



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$$W = F \Delta r \cos \theta = \vec{\mathbf{F}} \cdot \Delta \vec{\mathbf{r}}$$

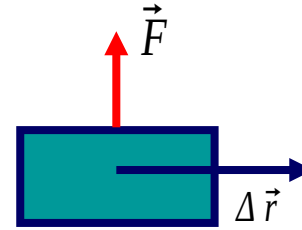
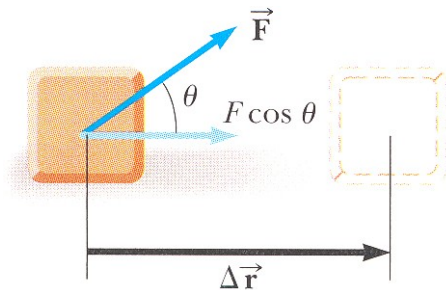


Some calculus.

Work = dot product force, displacement

- The work W done on a system by an agent exerting a constant force on the system is the product of the magnitude F of the force, the magnitude Δr of the displacement of the point of application of the force, and $\cos\theta$, where θ is the angle between the force and displacement vectors:

$$W \equiv \vec{F} \cdot \Delta \vec{r} = F \Delta r \cos \theta$$



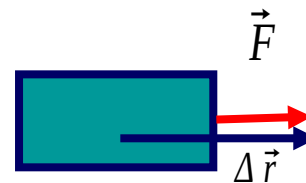
I

$$W_I = 0$$



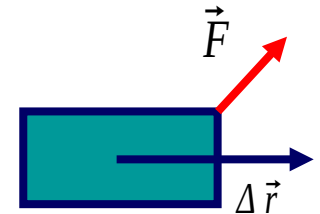
II

$$W_{II} = -F \Delta r$$



III

$$W_{III} = F \Delta r$$



IV

$$W_{IV} = F \Delta r \cos \theta$$



Scalar Product

- The vectors $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = -\hat{i} + 2\hat{j}$
- Determine the scalar product $\vec{A} \cdot \vec{B} = ?$
- Find the angle θ between these two vectors



Scalar Product

□ The vectors $\vec{A} = 2\hat{i} + 3\hat{j}$ and $\vec{B} = -\hat{i} + 2\hat{j}$

□ Determine the scalar product $\vec{A} \cdot \vec{B} = ?$

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y = 2 \cdot (-1) + 3 \cdot 2 = -2 + 6 = 4$$

□ Find the angle θ between these two vectors

$$A = \sqrt{A_x^2 + A_y^2} = \sqrt{2^2 + 3^2} = \sqrt{13} \quad B = \sqrt{B_x^2 + B_y^2} = \sqrt{(-1)^2 + 2^2} = \sqrt{5}$$

$$\cos \theta = \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{4}{\sqrt{13}\sqrt{5}} = \frac{4}{\sqrt{65}}$$

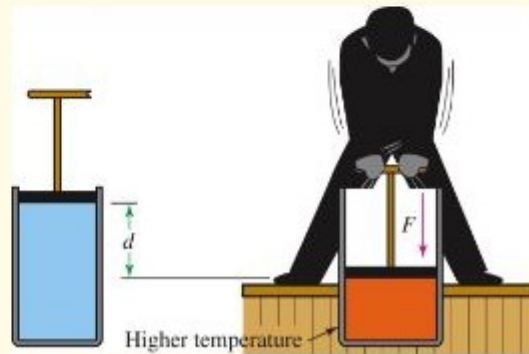
$$\theta = \cos^{-1} \frac{4}{\sqrt{65}} = 60.3^\circ$$

Textbook 1.9 1.10



First law of thermodynamics, cont'd

- Consider piston pushing on a gas in a cylinder.
 - Forcing the piston down requires a force.
 - Applying this force through a distance means you are doing work.
 - The work is done against the gas.
 - You compress the gas.
 - The gas gets hot.



Energy

■ **Energy** is defined as the measure of a system's ability to do work. That is to apply a force over a distance. This idea of energy was unknown to Newton. If an object has energy, damage can be done to another Object by going into it = transfer of energy

- We use the symbol E to represent energy.
- Energy has the same units as work:
 - Joule for SI, ft·lb for English

UNITS again

■ The units of energy:

■ Metric

- SI: **joule** ($J = N \cdot m$),
- erg ($= 10^{-7} J = 0.0000001$),
- calorie ($cal = 4.186 J$),
- 1 food Cal = 1000 cal = 4,186 J
- kilowatt-hour ($kWh = 0.278 J$).

■ English:

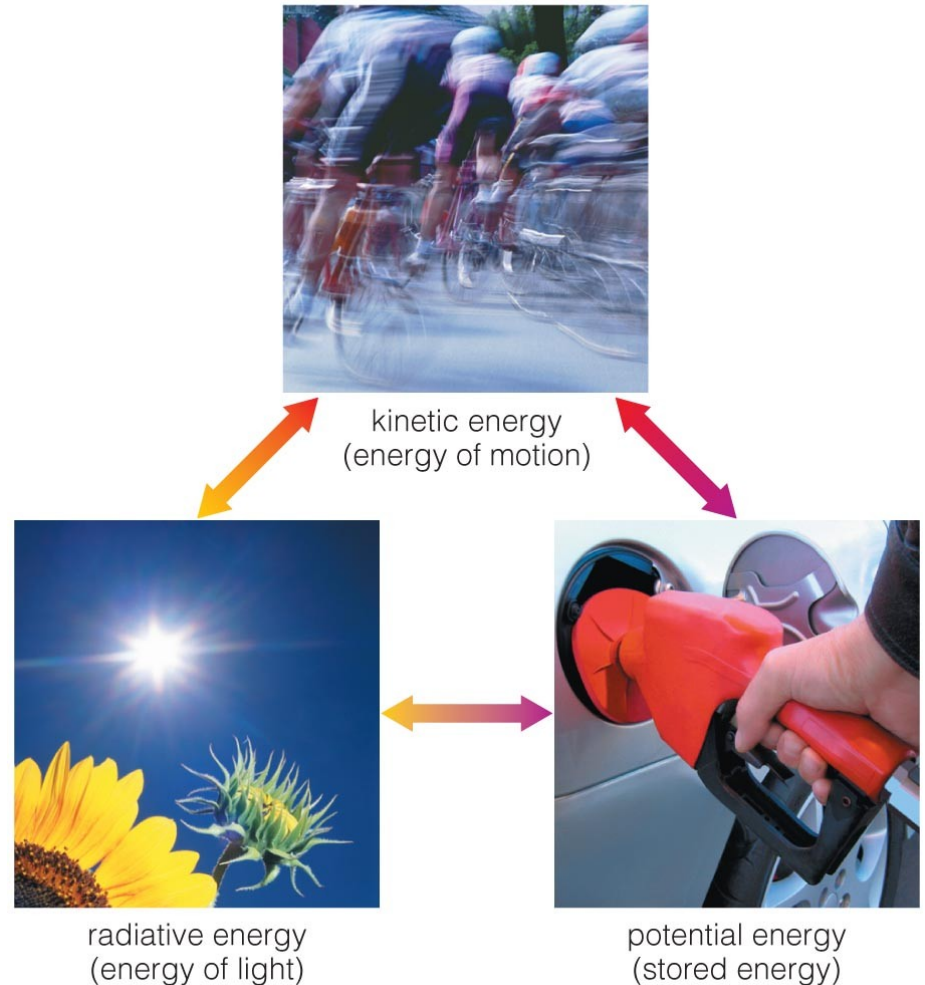
- foot-pound ($ft \cdot lb$),
- British thermal unit (Btu).

FORMS OF ENERGY

Kinetic (motion)
Radiative (light)
Potential (stored)

Energy can change type,
but cannot be created or
destroyed.

Energy can be converted from one form to another.



KINETIC ENERGY

- There are various types of energy.
 - **Kinetic energy** is the energy associated with an object's motion.

- We use the symbol KE . $KE = \frac{1}{2} m V^2$

■ **Work = Change in kinetic energy : $\Delta KE = 0.5 m (V_f^2 - V_i^2)$**

Note that the kinetic energy depends on the speed squared

So if you double your speed on the highway , damage during a potential Crash is multiply by 4 !!! This was not known to Newton.

Leibniz (co-founder of calculus) got the idea after an experiment was done with dropping balls in clay . Double the speed the balls have, Multiply the marks in the clay by 4 !!

Example:

1) A 50kg child runs at a speed of 5m/s and stop.
What is the change in kinetic energy ? HINT:

First find the Final kinetic energy
Then the initial kinetic energy then :

important

Change= FINAL – INITIAL

2) A person has a mass of 45kg and is moving with a velocity of 10m/s.
Find the person's kinetic energy?

The person's velocity becomes 5m/s. What is the kinetic energy ?

What is the ratio of the speeds ?
What is the ratio of the kinetic energy ?
Conclusion ?

In PHYSICS:

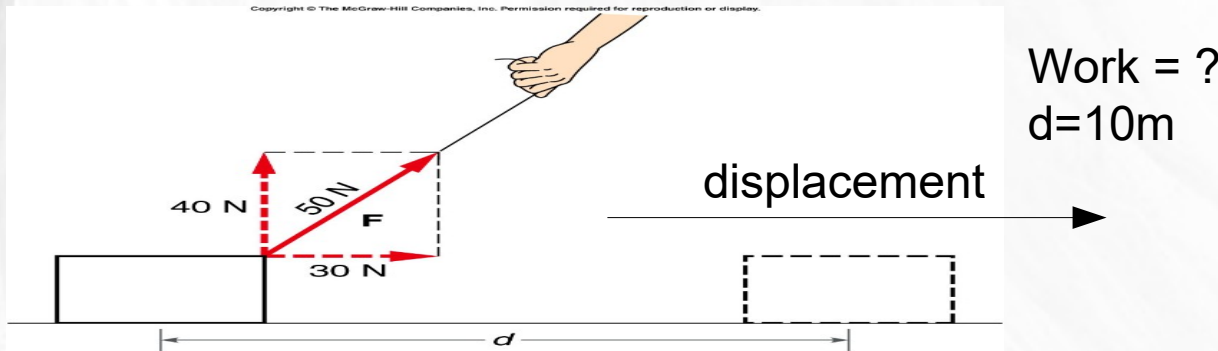
WORK MEASURES The AMOUNT of ENERGY

of motion Transferred FROM one SYSTEM to another. Work on a system is done by a force only if it changes the kinetic energy of the system.

Moving a mass along the x-axis - $\rightarrow$ work = $F_x \Delta X$

F_x is the x-component of the force - ΔX is the displacement along x-axis

Work in joules, distance in meter, force F in newtons



Example: You push a box with a force of 100N and if the box is moving by 4m Then the energy transferred from your Body (energy stored in fat) to the box (energy of motion)

= _____ x _____ = _____ J

Work, cont'd

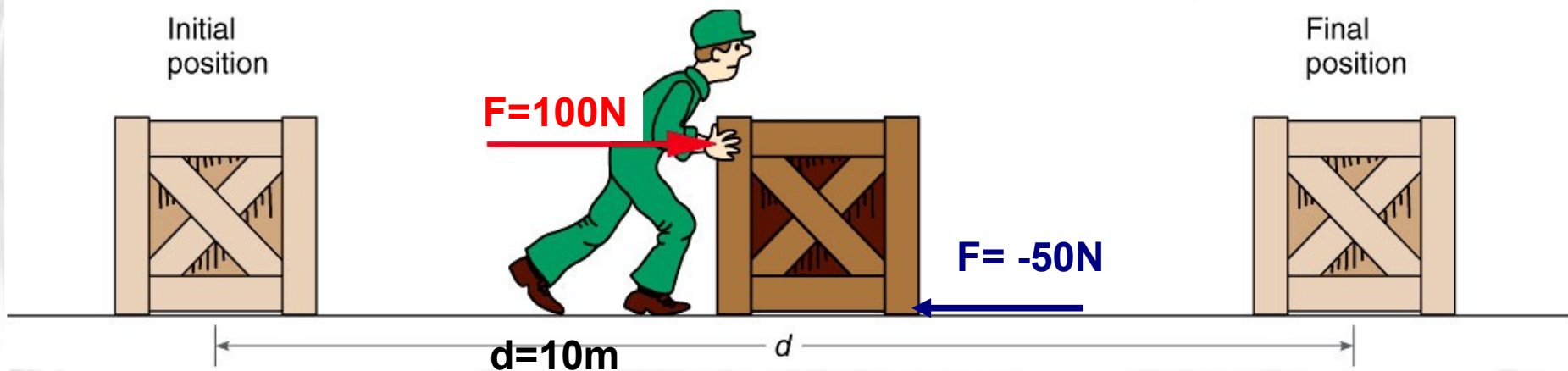
- Recall that force is a vector.
 - Involves magnitude and direction.
- Work is just that part of the force in the direction of the displacement.
 - Work is not a vector — it's a scalar.
- But the sign of the work does depend on the relative directions.

If work < 0 energy is removed from the system

- **If work > 0 energy is added to the system**
also: work is done on a system only if the speed changes.

- If the force and distance are in the same direction, the force does *positive work*.

If the force and distance are in the opposite direction, the force does *negative work*.



Work by the guy = ____ J work by the friction = ____ J
NET Work = ____ J CHANGE IN ENERGY = _____ J

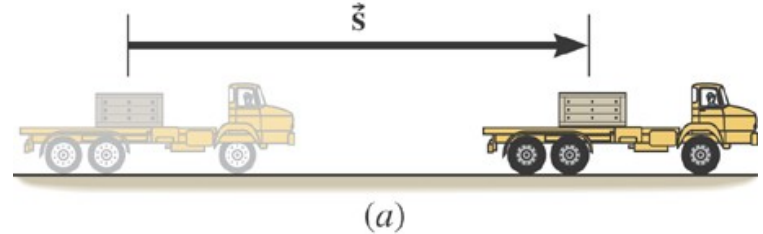
6.1 Work Done by a Constant Force

Example 3 Accelerating a Crate

The truck is accelerating at a rate of $+1.50 \text{ m/s}^2$. The mass of the crate is 120-kg and it does not slip. The magnitude of the displacement is 65 m.

What is the total work done on the crate by all of the forces acting on it?

We don't have the thrust and friction we have the net force $\rightarrow$ net work



The system is the
Crate. Draw a free-body diagram
For a crate. The crate has the
Same acceleration as the truck
Dah!



6.1 Work Done by a Constant Force

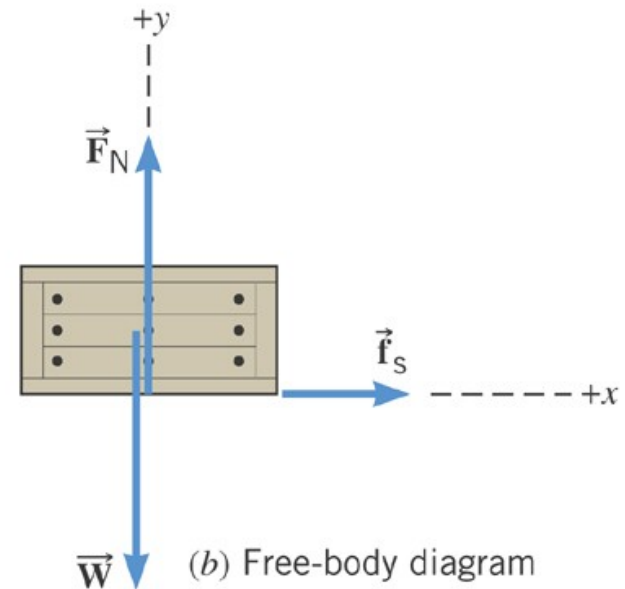
The angle between the displacement and the normal force is 90 degrees.

The angle between the displacement and the weight is also 90 degrees.

$$W = (F \cos 90) s = 0$$



(a)



(b) Free-body diagram for the crate



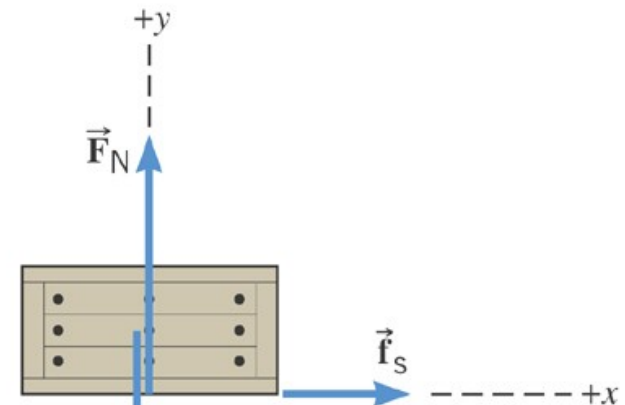
6.1 Work Done by a Constant Force



(a)

$$f_s = ma = (120 \text{ kg})(1.5 \text{ m/s}^2) = 180 \text{ N}$$

$$W = [(180 \text{ N}) \cos 0](65 \text{ m}) = 1.2 \times 10^4 \text{ J}$$



(b) Free-body diagram for the crate



suitcase has a mass of 10kg.

Find its weight in newtons

If you want to carry the suitcase, what is the Upward force you need to exert on it?

The upward force is balanced by the _____ of the suitcase.

Are you doing work on the suitcase?

Is gravity doing work?

No because they don't increase the speed
In the forward direction.

The forces are perpendicular to displacement.
Displacement. $W_{\text{done}} = 0$

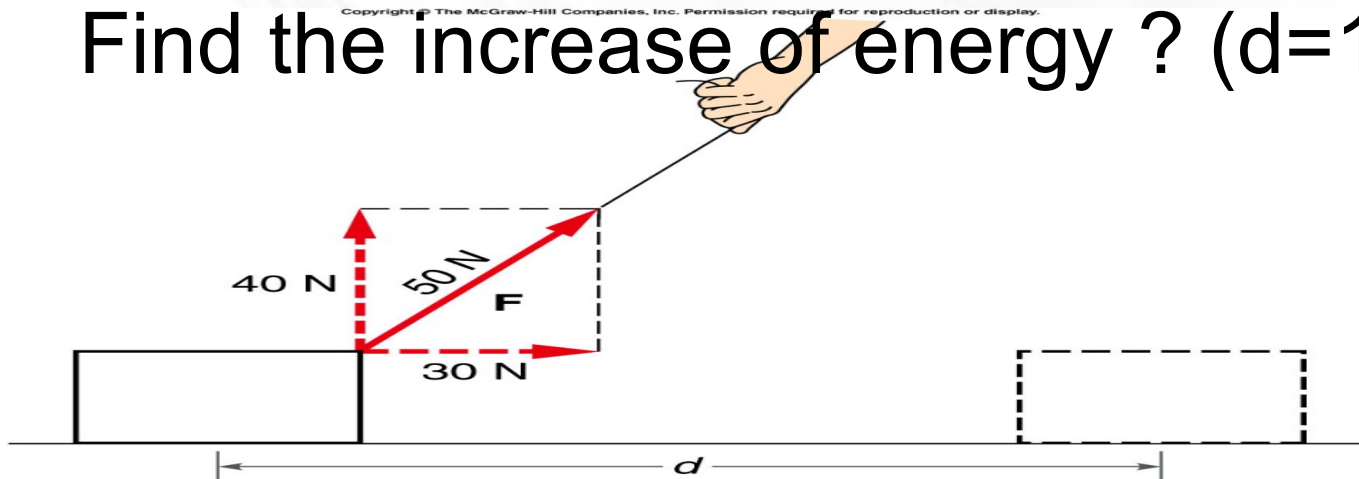


If the component of the force is not in the direction of the direction, the force does *no work*.

Bellow the pulling force for has 2 components, which component can increase the energy of the box (10kg) ?

So which component does work ?

Find the increase of energy ? ($d=10\text{m}$)



Work Done by Multiple Forces

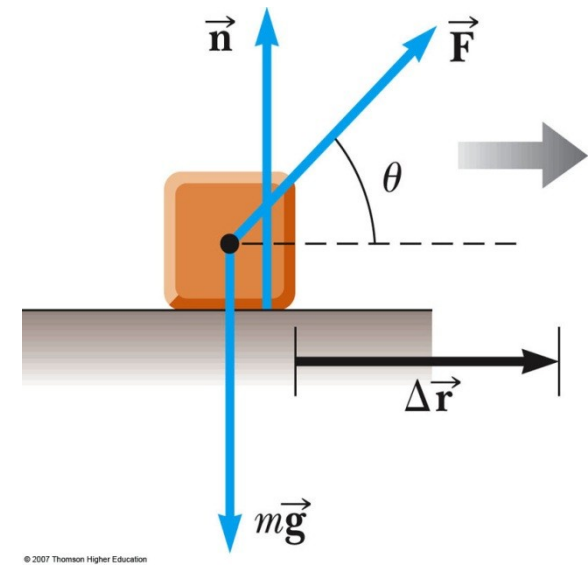
- If more than one force acts on an object, then the total work is equal to the algebraic sum of the work done by the individual forces

$$W_{\text{net}} = \sum W_{\text{by individual forces}}$$

- Remember work is a scalar, so this is the algebraic sum

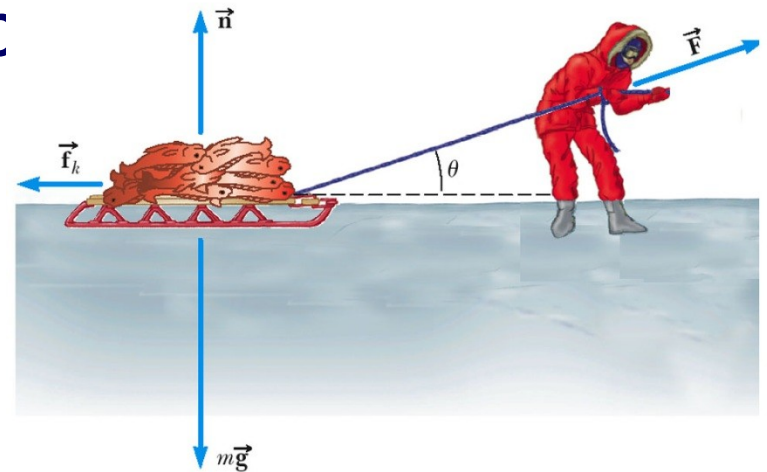
$$W_{\text{net}} = W_g + W_N + W_F = (F \cos \theta) \Delta r$$

Textbook 6.2



Work and Force

- ❑ An Eskimo pulls a sled as shown. The total mass of the sled is 50.0 kg, and he exerts a force of 1.20×10^2 N on the sled by pulling on the rope. How much work does he do on the sled if $\theta = 30^\circ$ and he pulls the sled 5.0 m ?
- ❑ If $\mu_k = 0.1$ what's the work W_f ?
- ❑ What's the net work ?



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Alternative way: find first
The net force and then
The net work



Positive or negative work?
positive work = increase of energy of the system. you are working against gravity even if work is negative.

$$W = (F \cos 0) s = Fs$$

$$W = (F \cos 180) s = -Fs$$

On way up, you are doing positive work

Gravity is doing negative work (show)

On way down you are doing negative work

Gravity is doing positive work

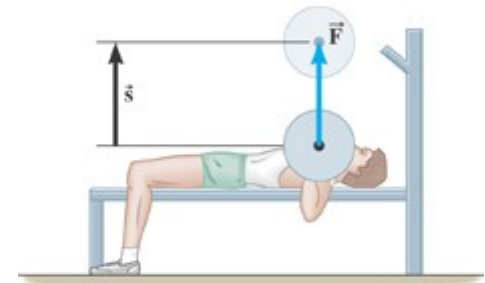
Say on way up he does 500J of work

Lifting 50kg. What his work done on dumbbell?

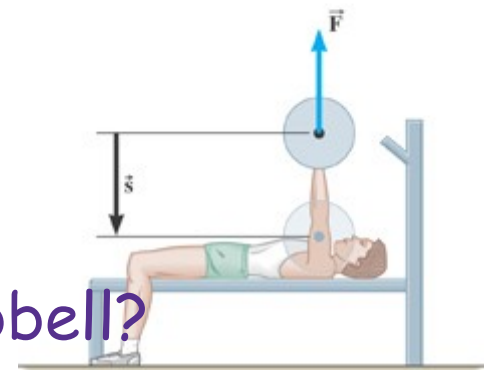
What is the net work? Why?



(a)

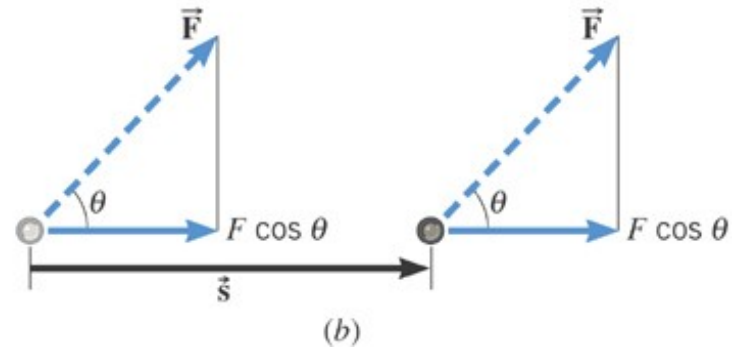
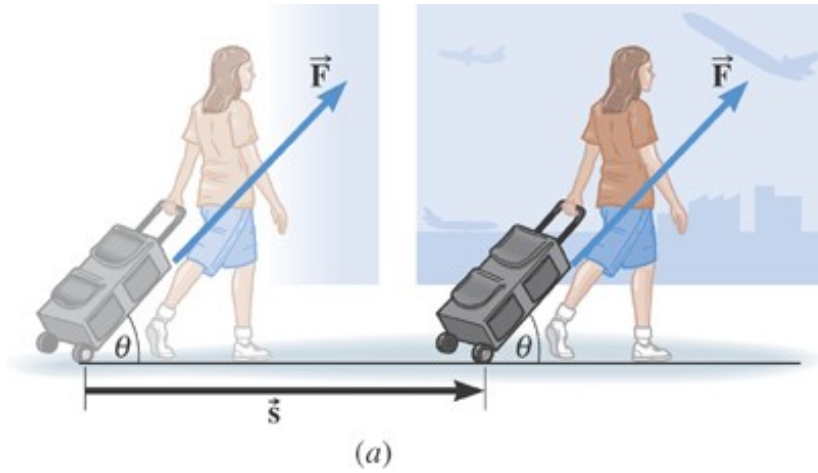


(b)



(c)





Textbook 7.1 and 7.3

Work-Kinetic Energy Theorem

When work is done by a net force on an object and the only change in the object is its speed, the work done is equal to the change in the object's kinetic energy

- Speed will increase if work is positive
- Speed will decrease if work is negative

Special case: Constant Acceleration

Remember result eliminating t :

$$v^2 - v_0^2 = 2a(x - x_0)$$

Multiply by $\frac{1}{2}m$:

$$\begin{aligned}\frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 &= ma(x - x_0) \\ &= ma\Delta x\end{aligned}$$

But
 $F=ma!$

$$\Delta\left(\frac{1}{2}mv^2\right) = F\Delta x$$



Example:

You are pushing a box with a force of 100N
But a 50N frictional force opposes the motion.
Distance is 10m

1) Find the work done by each force

2) Find the net work (distance is 5m)

3) Find the increase in kinetic energy

4) If the initial speed of the box is 0 (starts from rest) and if the Box is 50 kg, find the final speed.

$$KE = 0.5 m v^2$$

Example

Because of friction, a constant force of 100 newtons is needed to slide a box across a room. If the box moves 3 meters, how much work must be done? The speed is constant.

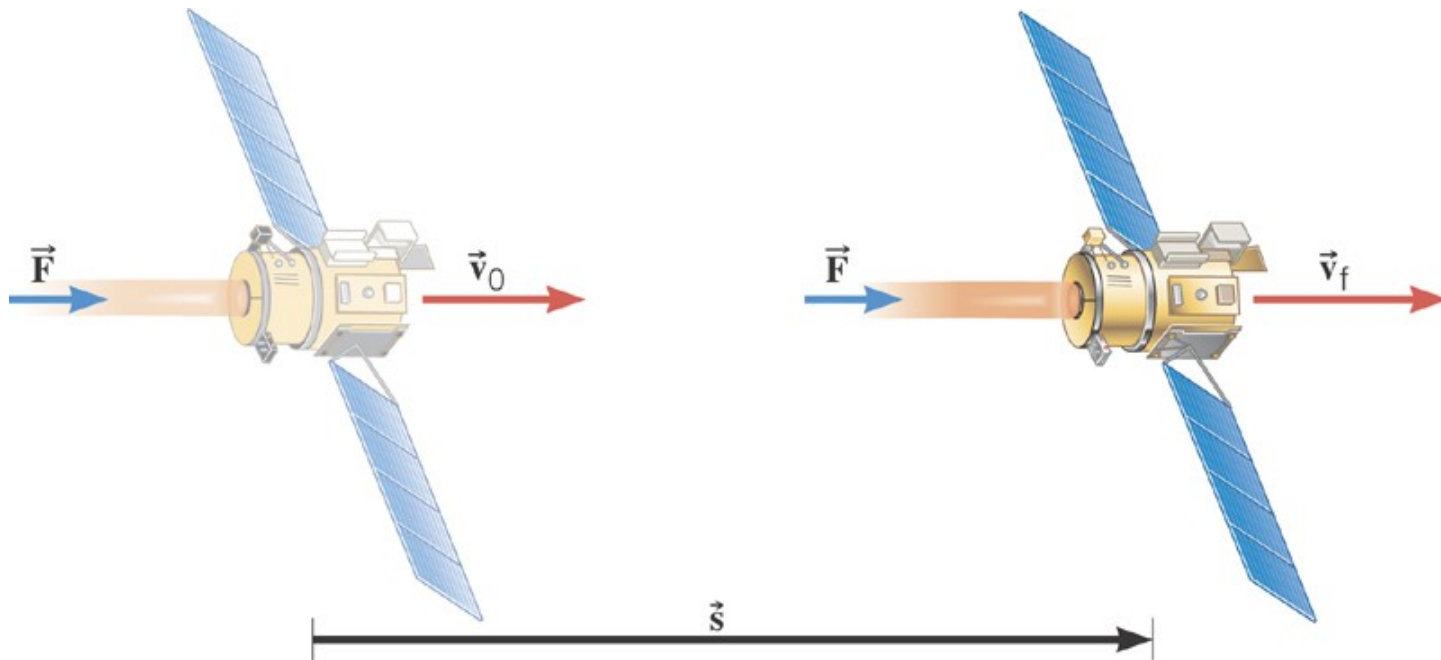
(that is how much work you are doing on the box).

If the friction is -100N , what is the work done by the Friction force ?

What is the total net force ? the net work ?
The change in kinetic energy ?

Example 4 Deep Space 1

The mass of the space probe is 474-kg and its initial velocity is 275 m/s. If the 0.056 N force acts on the probe through a displacement of 2.42×10^9 m, what is its final speed?

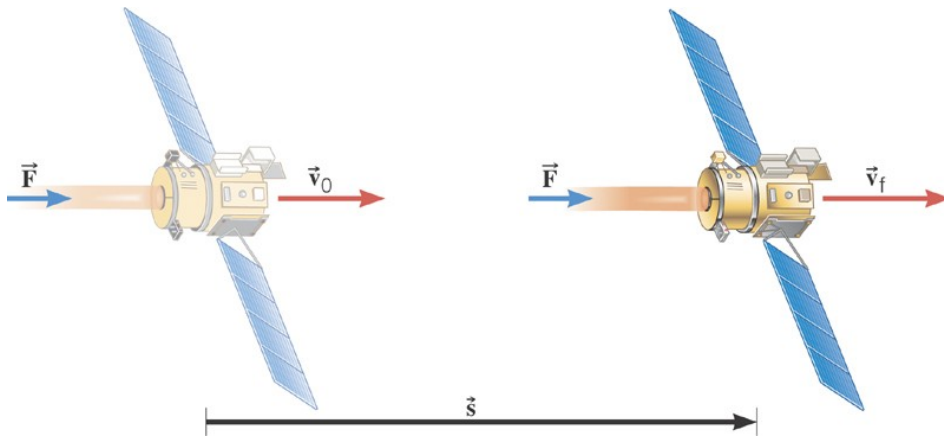


$$\left[\left(\sum F \right) \cos \theta \right] s = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_o^2$$

$$(5.60 \times 10^{-2} \text{ N}) \cos 0^\circ (2.42 \times 10^9 \text{ m}) = \frac{1}{2} (474 \text{ kg}) v_f^2 - \frac{1}{2} (474 \text{ kg}) (275 \text{ m/s})^2$$



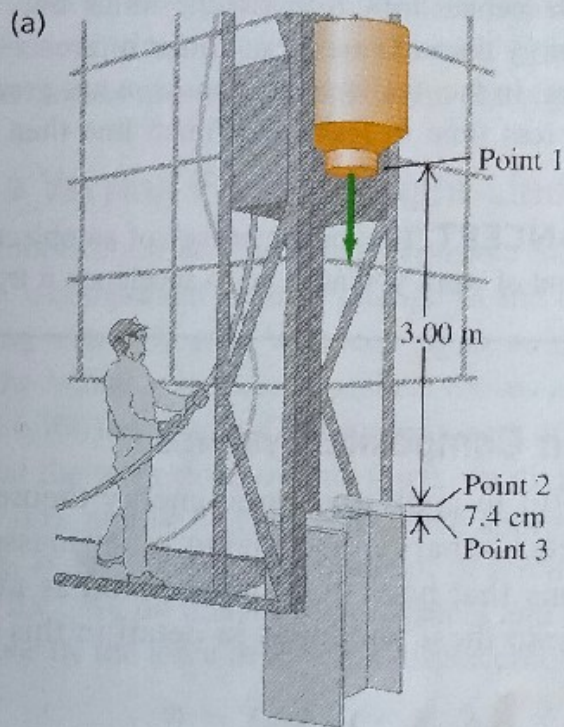
$$v_f = 805 \text{ m/s}$$



Problem 6.4

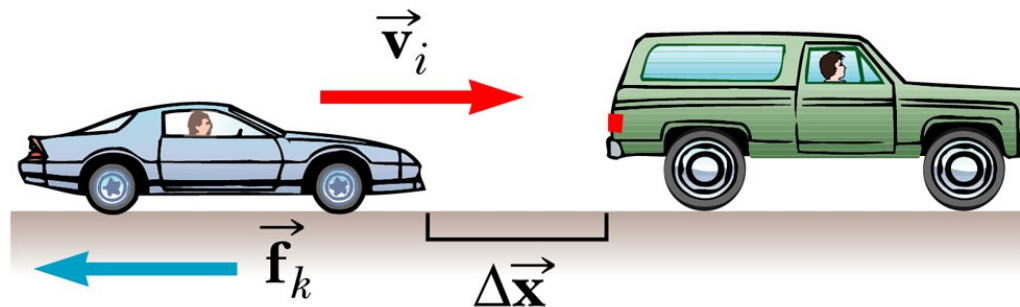
The 200 kg steel hammerhead of a pile driver is lifted 3.00 m above the top of a vertical I-beam being driven into the ground (**Fig. 6.12a**). The hammerhead is then dropped, driving the I-beam 7.4 cm deeper into the ground. The vertical guide rails exert a constant 60 N friction force on the hammerhead. Use the work–energy theorem to find (a) the speed of the hammerhead just as it hits the I-beam and (b) the average force the hammerhead exerts on the I-beam. Ignore the effects of the air.

Figure 6.12 (a) A pile driver pounds an I-beam into the ground.



Work and Kinetic Energy

- The driver of a 1.00×10^3 kg car traveling on the interstate at 35.0 m/s slam on his brakes to avoid hitting a second vehicle in front of him, which had come to rest because of congestion ahead. After the breaks are applied, a constant friction force of 8.00×10^3 N acts on the car. Ignore air resistance. (a) At what minimum distance should the brakes be applied to avoid a collision with the other vehicle? (b) If the distance between the vehicles is initially only 30.0 m, at what speed would the collisions occur?



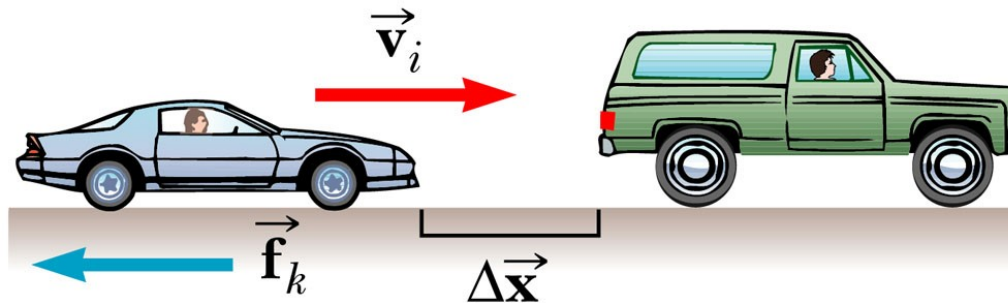
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Work and Kinetic Energy

- (a) We know $v_0 = 35.0 \text{ m/s}$, $v = 0$, $m = 1.00 \times 10^3 \text{ kg}$, $f_k = 8.00 \times 10^3 \text{ N}$
- Find the minimum necessary stopping distance

$$\begin{aligned} W_{\text{net}} &= W_{\text{fric}} + W_g + W_N = W_{\text{fric}} = \frac{1}{2}mv_f^2 - \frac{1}{2}mv_i^2 \\ -f_k \Delta x &= 0 - \frac{1}{2}mv_0^2 \\ -(8.00 \times 10^3 \text{ N}) \Delta x &= -\frac{1}{2}(1.00 \times 10^3 \text{ kg})(35.0 \text{ m/s})^2 \\ \Delta x &= 76.6 \text{ m} \end{aligned}$$



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Work and Kinetic Energy

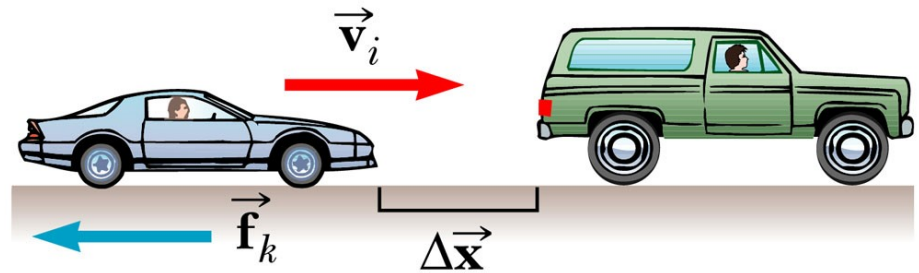
- (b) We know $\Delta x = 30.0 \text{ m}$, $v_0 = 35.0 \text{ m/s}$, $m = 1.00 \times 10^3 \text{ kg}$, $f_k = 8.00 \times 10^3 \text{ N}$
- Find the speed at impact.
- Write down the work-energy theorem:

$$W_{\text{net}} = W_{\text{fric}} = -f_k \Delta x = \frac{1}{2} m v_f^2 - \frac{1}{2} m v_i^2$$

$$v_f^2 = v_0^2 - \frac{2}{m} f_k \Delta x$$

$$v_f^2 = (35 \text{ m/s})^2 - \left(\frac{2}{1.00 \times 10^3 \text{ kg}} \right) (8.00 \times 10^3 \text{ N}) (30 \text{ m}) = 745 \text{ m}^2/\text{s}^2$$

$$v_f = 27.3 \text{ m/s}$$

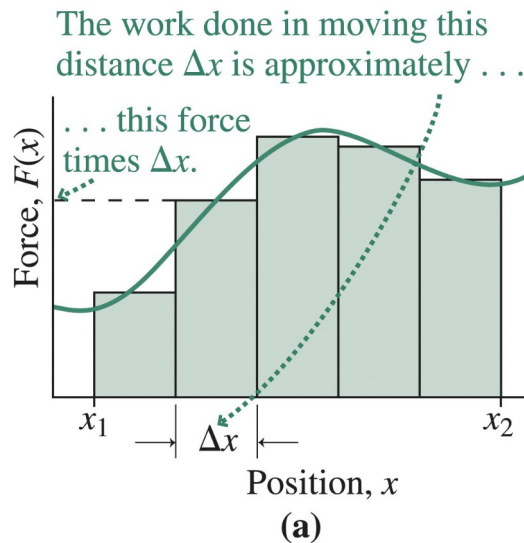


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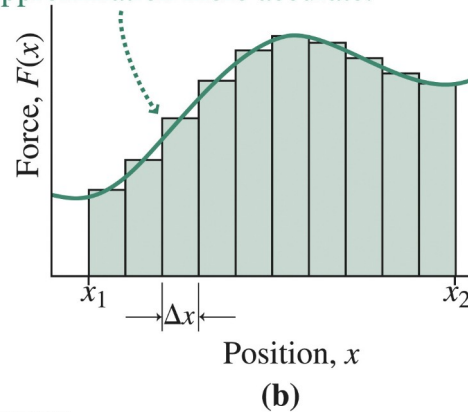


Work Done by a Varying Force

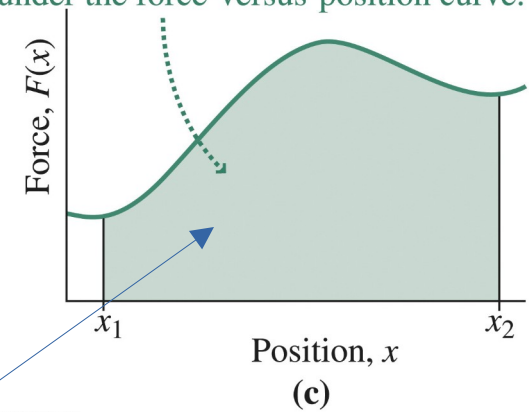
- When a force varies with position, it's necessary to integrate to calculate the work done.
- Geometrically, the work is the area under the force-versus-position curve.



Making the rectangles smaller makes the approximation more accurate.



The exact value for the work is the area under the force-versus-position curve.



She increases her pull, then gets tired
Then pulls hard again/



6.9 Work Done by a Variable Force

Constant Force

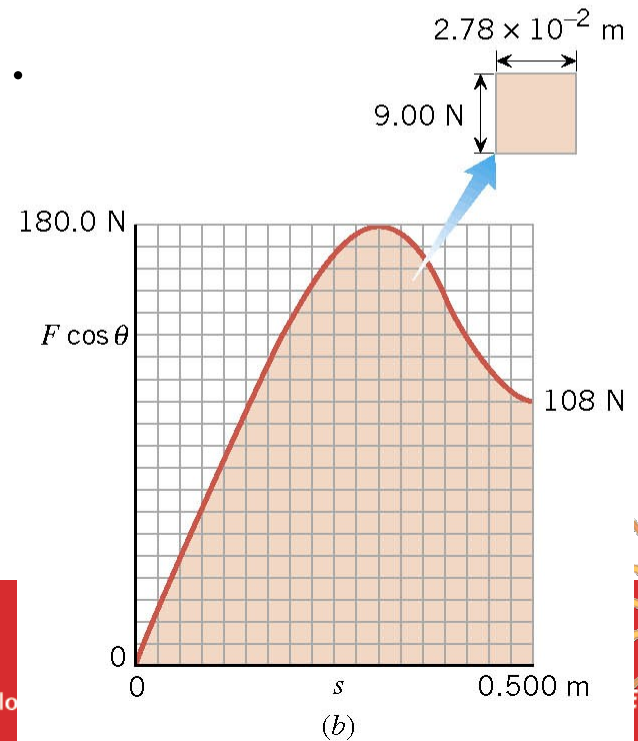
$$W = (F \cos \theta)s$$

Variable Force

$$W \approx (F \cos \theta)_1 \Delta s_1 + (F \cos \theta)_2 \Delta s_2 + \dots$$



(a)



Integration

- The *definite integral* is the result of the limiting process in which the area is divided into ever smaller regions.
- Work as the integral of the force F over position x is written

$$W = \int_{x_1}^{x_2} F(x) dx$$

Its the area under
Under the curve.

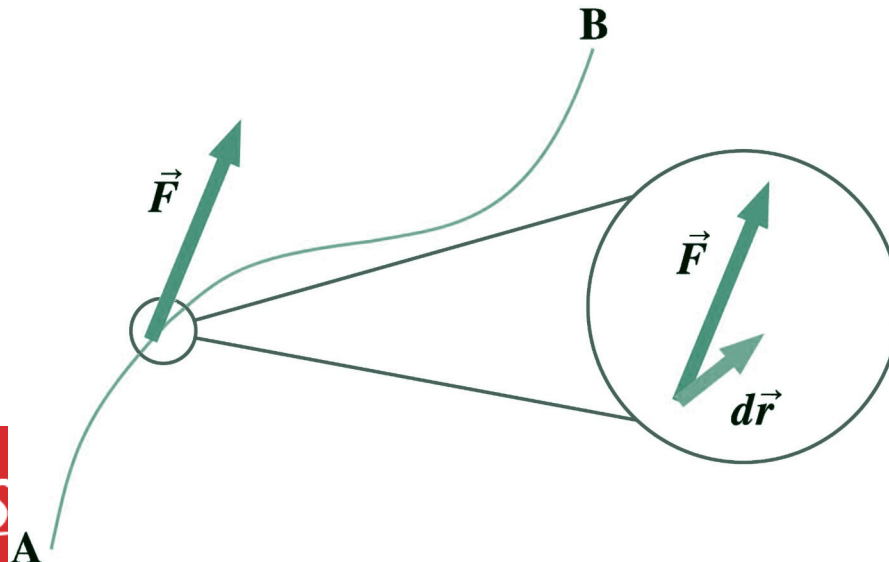
- Integration is the opposite of differentiation, so integrals of simple functions are readily evaluated. For powers of x , the integral becomes

$$\int_{x_1}^{x_2} x^n dx = \frac{x^{n+1}}{n+1} \Big|_{x_1}^{x_2} = \frac{x_2^{n+1}}{n+1} - \frac{x_1^{n+1}}{n+1}$$



A Varying Force in Multiple Dimensions

- In the most general case, an object moves on an arbitrary path subject to a force whose magnitude and whose direction relative to the path may vary with position.
- In that case the integral for the work becomes a **line integral**, the limit of the sum of scalar products of infinitesimally small displacements with the force at each point.



$$W = \int_{\vec{r}_1}^{\vec{r}_2} \vec{F} \cdot d\vec{r}$$



A spoon of calculus

The Work-Energy Theorem

- Applying Newton's second law to the net work done on an object results in the **work-energy theorem**:

$$W_{\text{net}} = \int F_{\text{net}} dx = \int ma dx = \int m \frac{dv}{dt} dx = \int m \frac{dx}{dt} dv = \int mv dv$$

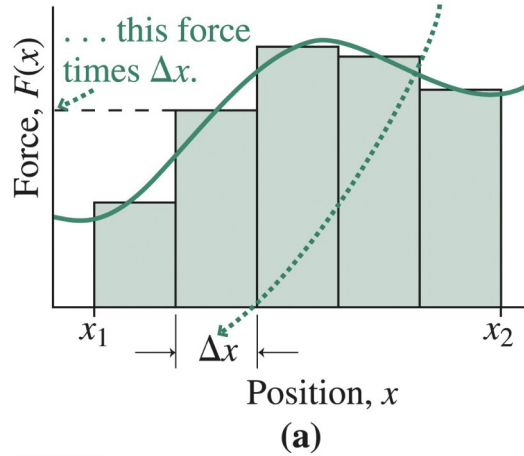
- Evaluating the last integral between initial and final velocities v_1 and v_2 gives

$$W_{\text{net}} = \int_{v_1}^{v_2} mv dv = \frac{1}{2} mv^2 \Big|_{v_1}^{v_2} = \frac{1}{2} mv_2^2 - \frac{1}{2} mv_1^2$$

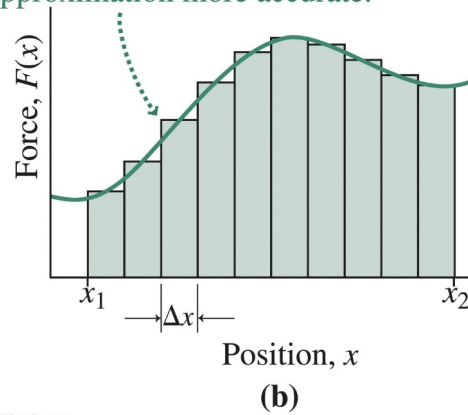
- So the quantity $\frac{1}{2} mv^2$ changes only when net work is done on an object, and the change in this quantity is equal to the net work.



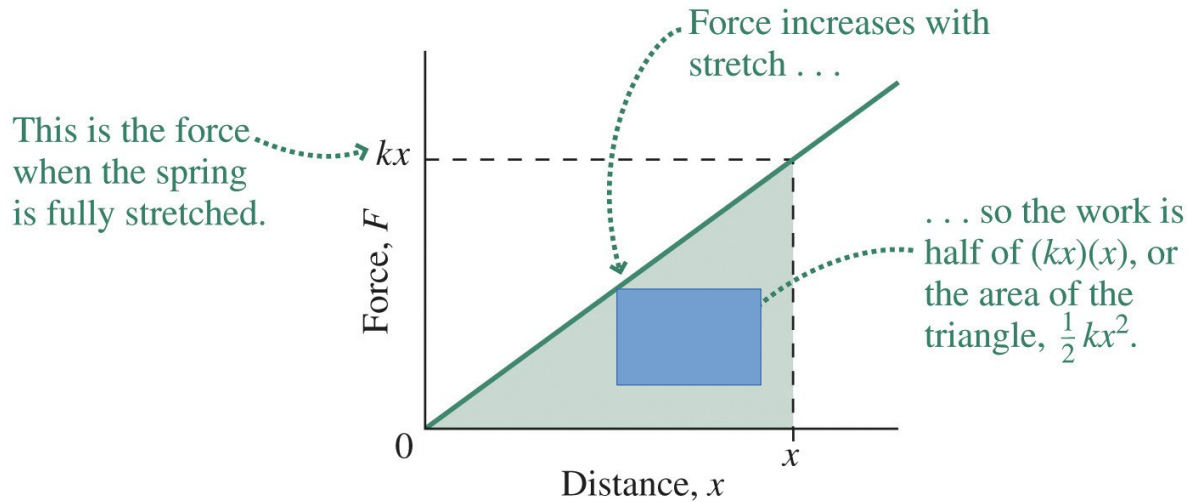
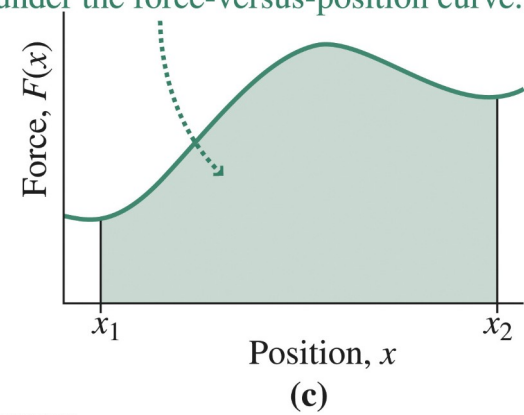
The work done in moving this distance Δx is approximately ...



Making the rectangles smaller makes the approximation more accurate.



The exact value for the work is the area under the force-versus-position curve.



What, is the area below
The graph $F = kx$?

Triangle $\rightarrow \frac{1}{2} (\text{base}) (\text{height})$



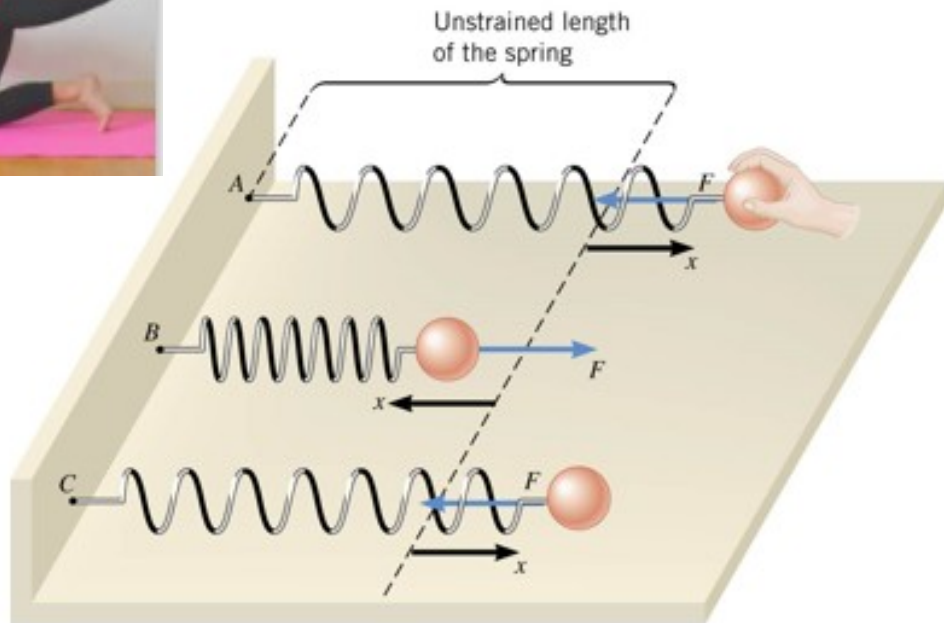
Case study → elastic PE - How to compute the PE

HOOKE'S LAW: RESTORING FORCE OF AN IDEAL SPRING

The restoring force on an ideal spring is $F_x = -kx$



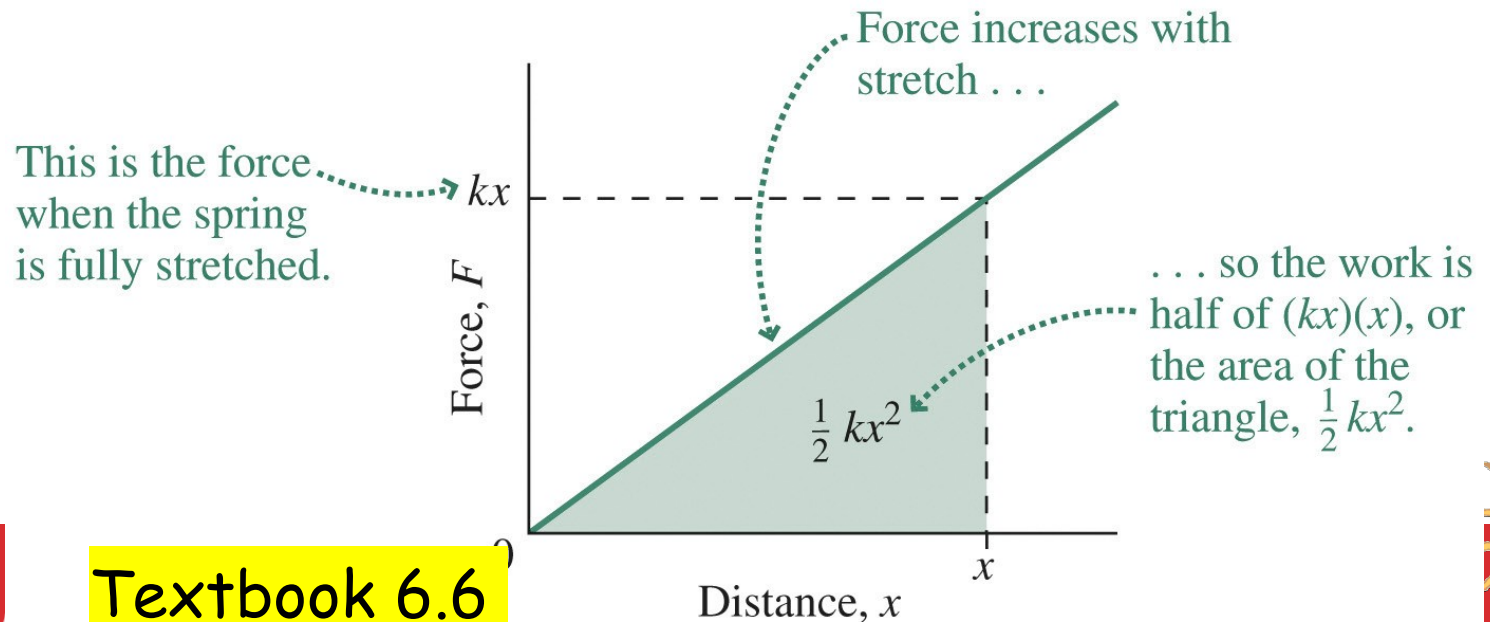
See movie
Walter Lewin



OR use calculus

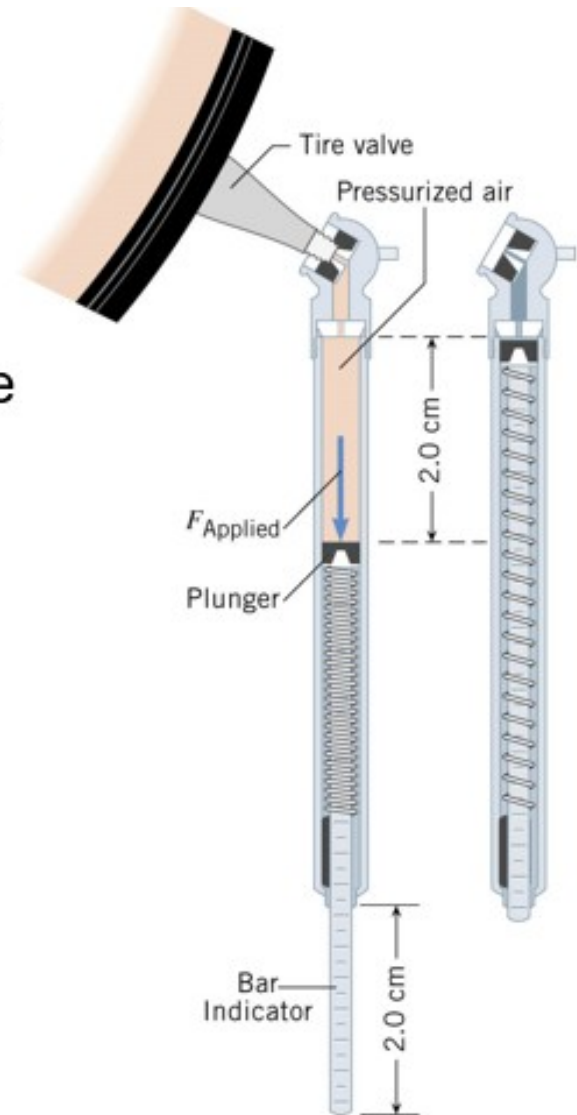
- A spring exerts a force $F_{\text{spring}} = -kx$.
- The agent stretching a spring exerts a force $F = +kx$, and the work the agent does is

$$W = \int_0^x F(x) dx = \int_0^x kx dx = \frac{1}{2}kx^2 \Big|_0^x = \frac{1}{2}kx^2 - \frac{1}{2}k(0)^2 = \frac{1}{2}kx^2$$



Example 1 A Tire Pressure Gauge

The spring constant of the spring is 320 N/m and the bar indicator extends 2.0 cm . What force does the air in the tire apply to the spring?



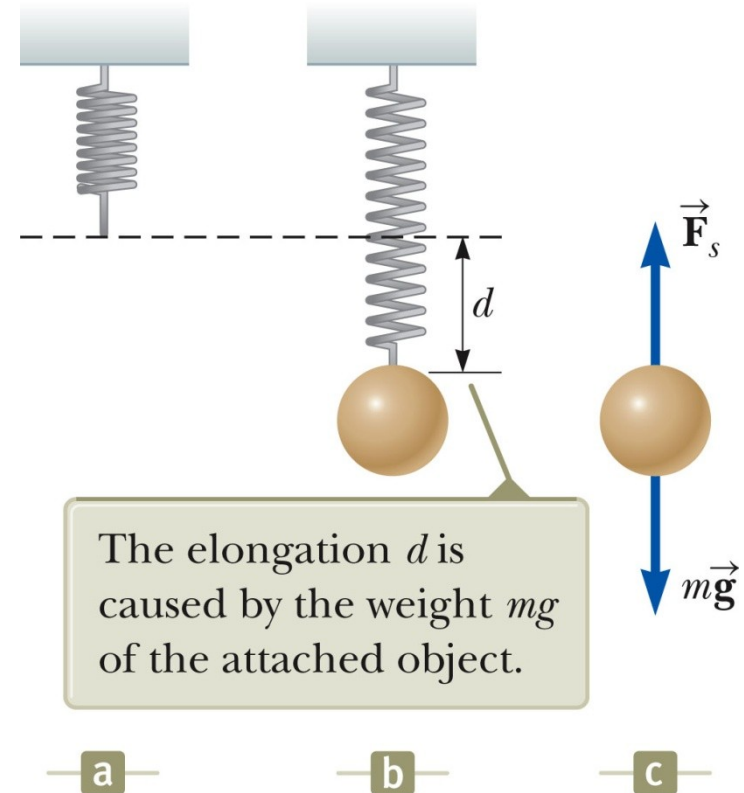
Measuring Spring Constant

- Start with spring at its natural equilibrium length.
- Hang a mass on spring and let it hang to distance d (stationary)

- From $F_x = kx - mg = 0$

$$k = \frac{mg}{d}$$

so can get spring constant.



https://energyeducation.ca/encyclopedia/Elastic_potential_energy

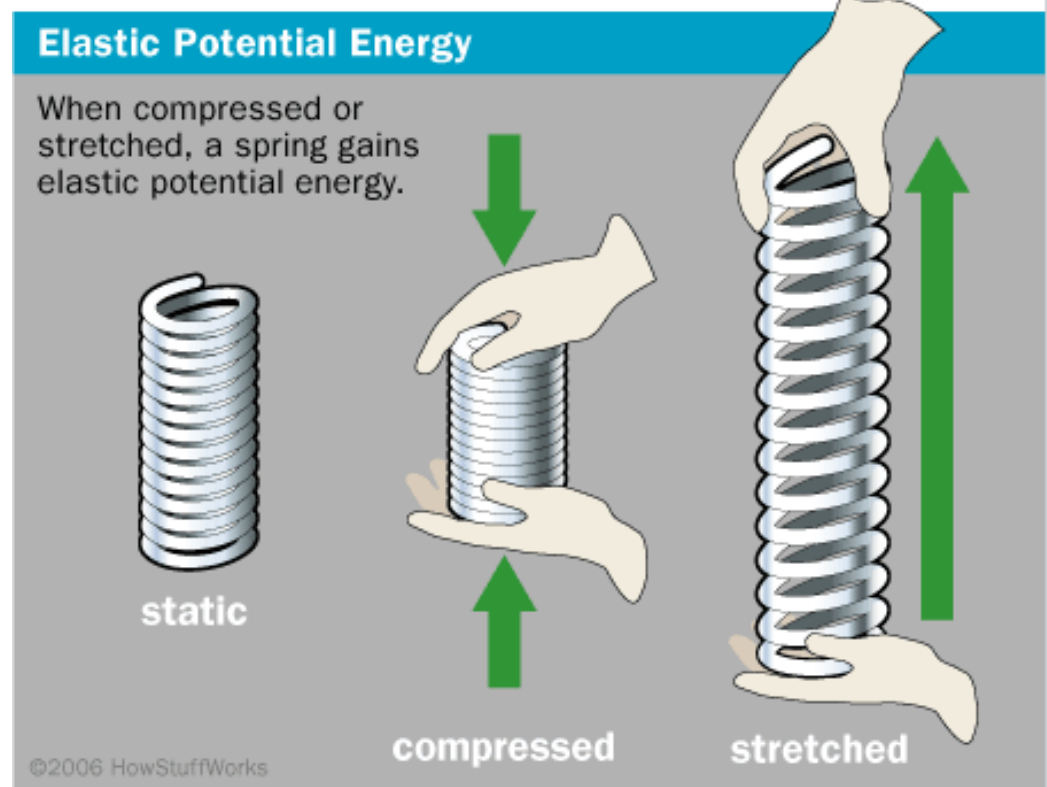


Figure 1. Elastic potential energy stored by a spring.^[1]

Elastic potential energy is given by the equation:^[3]

$$E_{\text{elastic}} = \frac{1}{2} kx^2 \text{ where}$$

- E_{elastic} : elastic potential energy (Joules, J)
- k : spring constant (Newtons per meter, N/m)
- x : distance from original (equilibrium) position (meters, m)

The elastic properties of a spring depend on both the shape and the material of the spring. Therefore, the spring constant is different for every object.

What about potential energy ? Discussed down below

POTENTIAL ENERGY IS STORED ENERGY. It's a binding energy. Or negative energy. Potential energy is energy associated with the **system's position or orientation**. We use the symbol *PE*. POTENTIAL ENERGY has always a restoring force (conservative force) That tries to bring the system to a lower level of energy/ Only the final and Initial positions matter. The restoring force is conservative.

- **CHEMICAL POTENTIAL ENERGY** (plants, coal, oil, cookies ..)

The energy is in the bonds between atoms. These forces Are like compressed springs. The can expand suddenly.

The forces are electromagnetic forces. (burning fuel, TNT ..)

- **NUCLEAR ENERGY** = energy stored in the nuclei. $E = mc^2$

Energy stored in the bonds between nucleons.

- **GRAVITATIONAL POTENTIAL ENERGY** (a book above the ground)

Gravity glue us to Earth, Earth to the Sun, the Sun to the galaxy..)

$\Delta PE = m g \Delta h$ ($g=10\text{m/s/s}$) close to Earth $\rightarrow$ PE is 0 at a reference level

- **ELASTIC POTENTIAL ENERGY**
(a compressed or stretched spring..)

$$\Delta PE = \frac{1}{2} K (\Delta x)^2 \quad \text{simulation}$$



1) You have a spring-loaded air rifle. When it is loaded, the spring is compressed 0.3m and has a spring constant of $K = 150\text{N/m}$. In joules, how much elastic potential energy is stored in the spring? Hint: elastic $PE = 0.5 K (X)^2$ X is the distance compressed or stressed.

2) How much work is required to compress a spring, $k = 33.3\text{N/m}$, from its equilibrium point at $x = 0.0\text{m}$ to $x = 5.0\text{m}$? (work = change in energy).

3) How much work is required to compress a spring ($k = 33.3\text{N/m}$) from $x = 5.0\text{m}$ to $x = 7.00\text{m}$?

A spring has a spring constant of 100 N/m. How much potential energy does it store when stretched by 10 cm?

The restoring force is $-KX$ (X displacement)

A restoring force derives from a PE (any)

Take the derivative of PE

What do you get?

So $F_{\text{restoring}} = -dPE/dx$ (if displacement along x)

We always compute the change in PE $\rightarrow$ Depends
On the position of a system relative to a reference level

The restoring force is said to be conservative.
The change in PE only depends on final and initial position.



Potential Energy (near Earth)

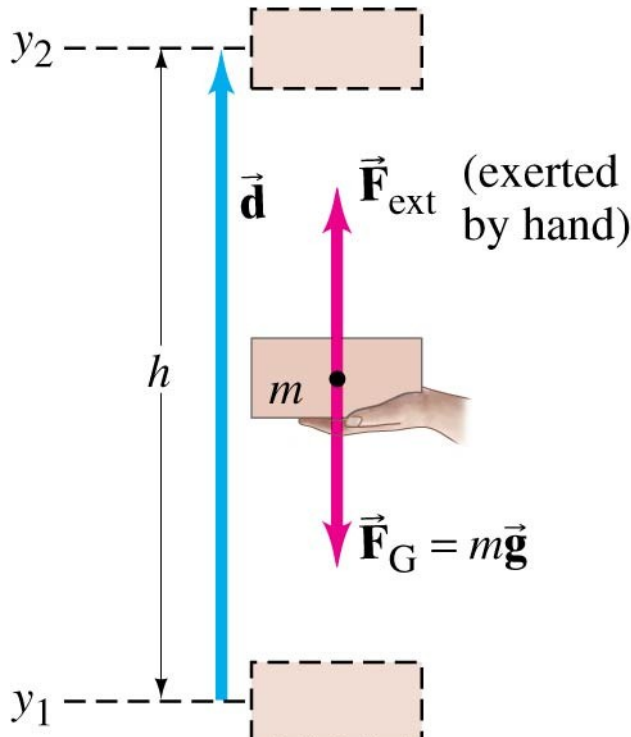
You need to specify a reference level

In raising a mass m to a height h , the work done by the external force is

$$W_{\text{ext}} = F_{\text{ext}} d \cos 0^\circ = mgh$$
$$= mg(y_2 - y_1)$$

we therefore define the gravitational potential energy:

$$\text{PE}_G = mgy.$$



Take the (-) of the derivative of PE_G . What do you get?
This is the restoring force. The change in PE depends only on initial position and on final position (conservative force).

Example

A 3-kg brick is lifted to a height of 0.5 meters above a table that is 1.0-m tall. Find the gravitational potential energy relative to the table and the floor.

$$\Delta PE = m g \times \Delta \text{height}$$

Potential energy is relative to a reference level
You can take $PE = 0$ where ever it is convenient
For computations. $PE=0$ at ground level.

The PE is meaningless without specifying the reference level.

Clicker Question 8



- To get to the second floor of the physics building, student A takes the elevator straight up, while student B walks up the stairway. Assuming that both students have the same mass, which of the following statements is true about their gravitational potential energy after reaching the second floor?
 - A. Student A has greater potential energy because he gets to the second floor faster.
 - B. Student B has greater potential energy because he has traveled a greater distance.
 - C. Both students have the same potential energy.
 - D. It is impossible to tell because potential energy depends on the time elapsed and distance traveled.



Two Kinds of Forces

Conservative forces

- “Give back” work done
 - Gravity
 - Electric force
 - Elastic (spring) forces (approximately)
- Work done is stored as *potential energy*

Nonconservative forces

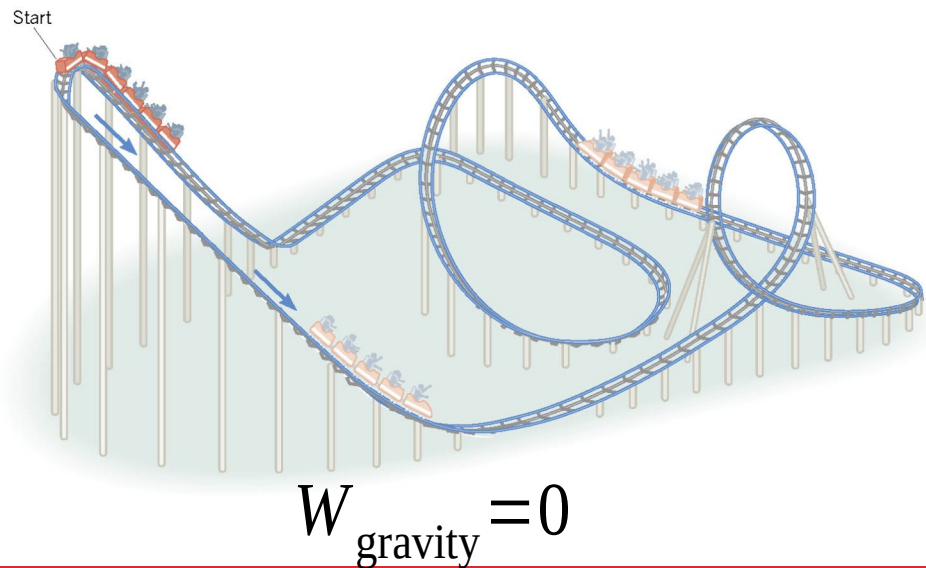
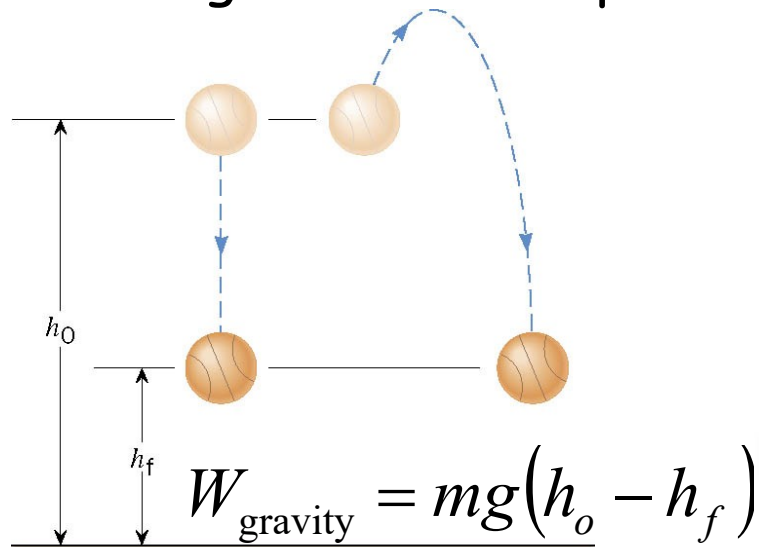
- Don’t “give back” work
 - Friction
 - Air resistance
 - Fluid viscosity
- Produce “heat”



→ All the forces that derive from a PE are conservative force

Version 1 A force is conservative when the work it does on a moving object is independent of the path between the object's initial and final positions.

Version 2 A force is conservative when it does no work on an object moving around a closed path, starting and finishing at the same point.



!! The work done by friction is not 0 if displacement is 0
Show why



- The “stored work” associated with a conservative force is called **potential energy**.
 - Potential energy is stored energy that can be released as kinetic energy.
- The change in potential energy is defined as the negative of the work done by a conservative force acting over any path between two points:

$$\Delta U_{AB} = -\int_A^B \vec{F} \cdot d\vec{r}$$

- Potential energy change is independent of path.
- Only *changes* in potential energy matter.
- We’re free to set the zero of potential energy at any convenient point.

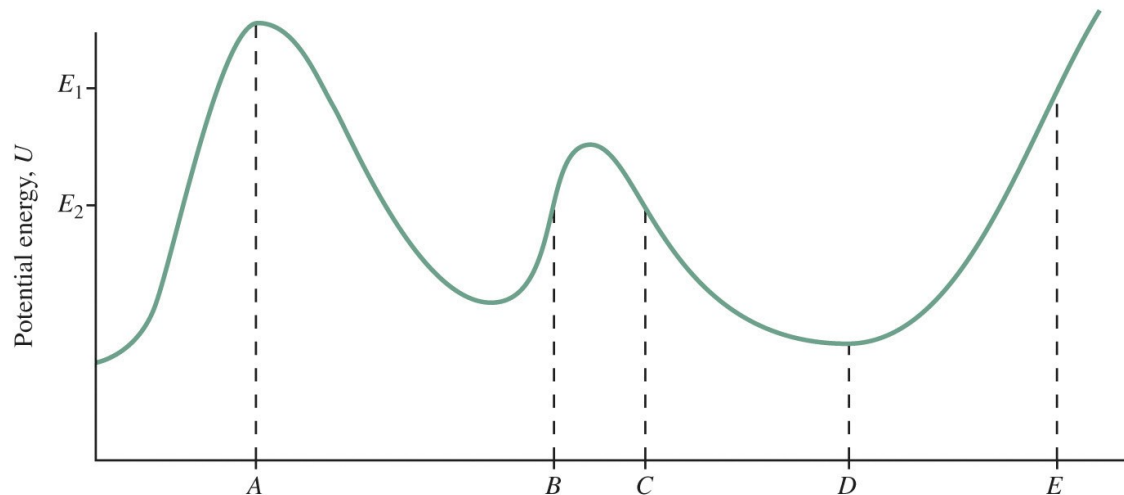
The force derive from a potential



Clicker Question 5



The figure shows the potential energy associated with an electron in a microelectronic device. From among the labeled points, find (1) the point where the force on the electron is greatest, (2) the right most position possible if the electron has total energy E_1 , (3) the leftmost position possible if the electron has total energy E_2 and starts out to the right of D, (4) a point where the force on the electron is zero, and (5) a point where the force on the electron points to the left. In some cases there may be more than one answer.



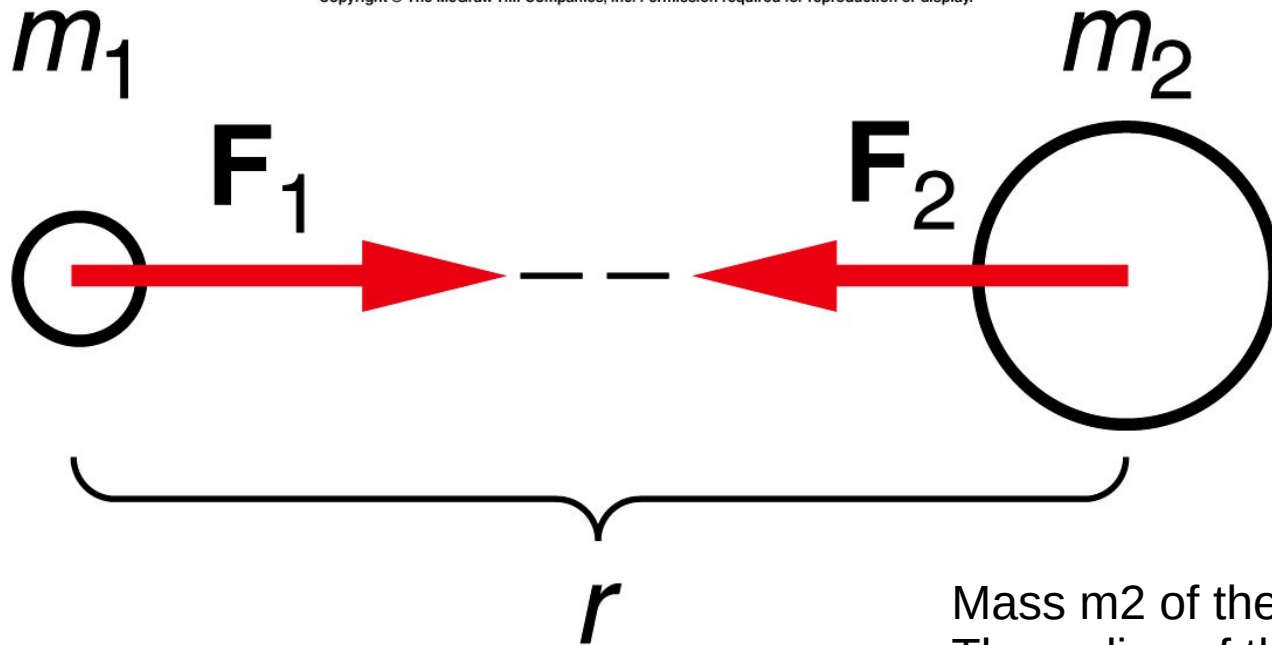
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Gravitational potential energy

Newton unified weight, motion on earth and motion in the heaven With his universal law of gravitation.

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$$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$$

$$F = G \frac{m_1 m_2}{r^2}$$

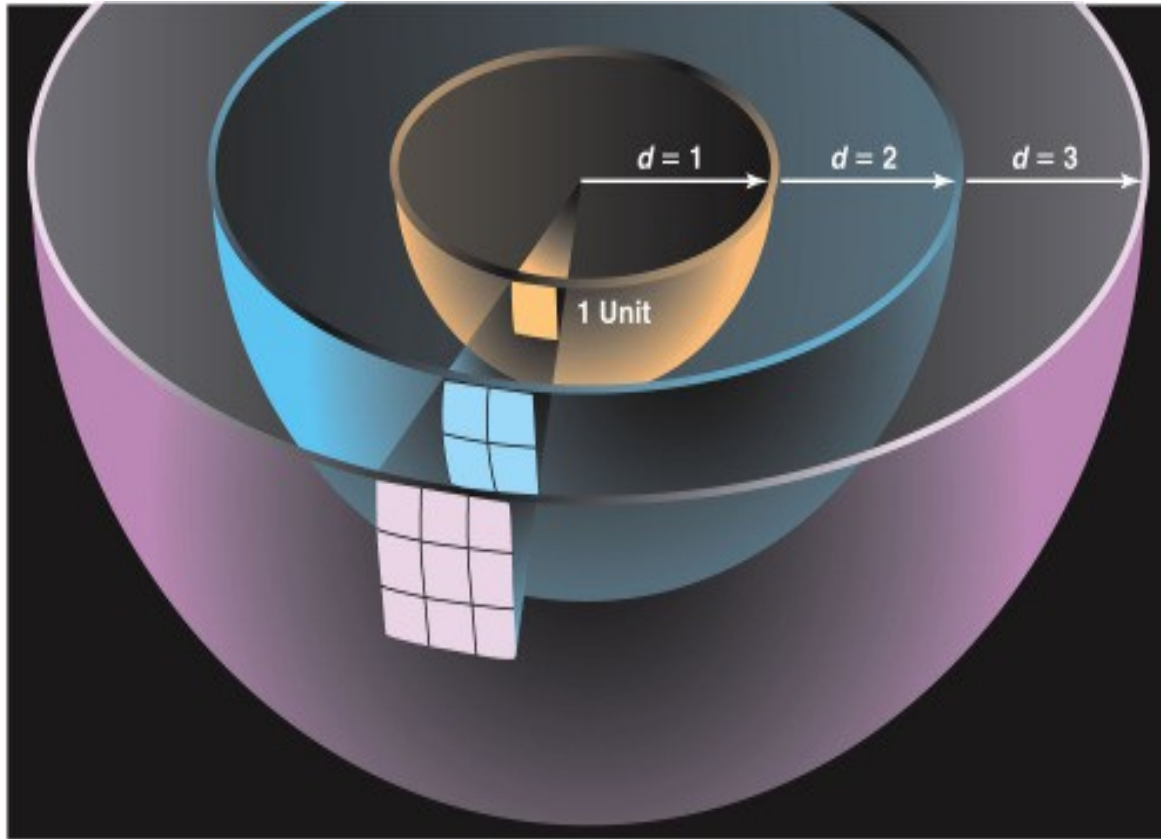
Mass m_2 of the Earth is 5.972×10^{24}
The radius of the Earth is 6,371 km
If you mass m_1 is 60kg:
A) find the pull due to gravity
B) compare to your weight.

30, 2021

6 and 6.7



Gravity is an inverse square law – lab inverse square law



Take the derivative
Of PE_g .

What do you get?

$PE = \text{at infinity}$

If you were to compress the Earth to the size of the Moon. The factor is 4.
What will happen to your weight? See exploration of physical sciences. gravity



Gravitation Potential Energy

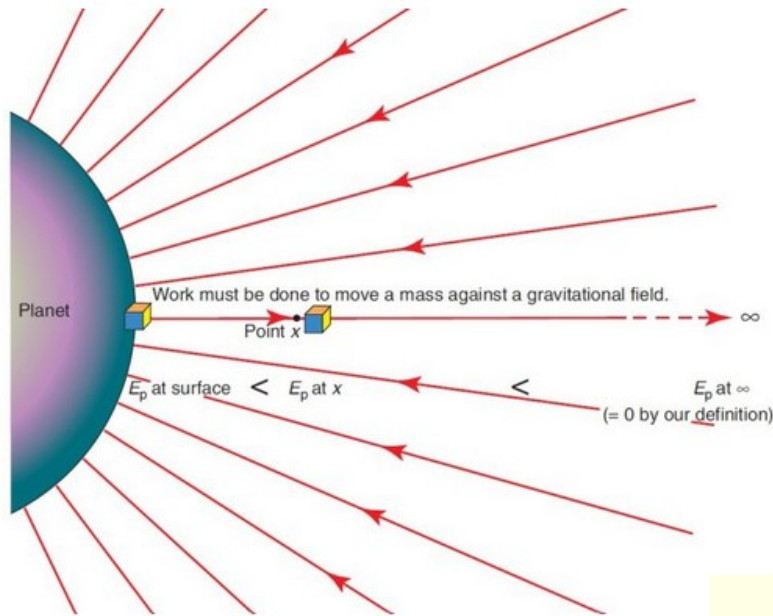


figure 2: gravitational potential energy E_p for a planet

Close to Earth we can approximate $U(r)$ with a line $\Delta U = mg h$ $h=r-R$

Its a linear approximation. (first degree)

Remember calculus:

$$f(x) - f(x_0) = f'(x=x_0) (x-x_0)$$

$$\Delta U = U'(r=R) (r-R) \text{ so } \Delta U = U'(r=R) h$$

$$\Delta U = mg h \text{ with } g = GM/R^2$$

Then you pick $U=0$ wherever you want

Gravitational Potential

From the definition of potential energy $U = -\int_{\text{ref}}^r \vec{F} \cdot d\vec{r}$

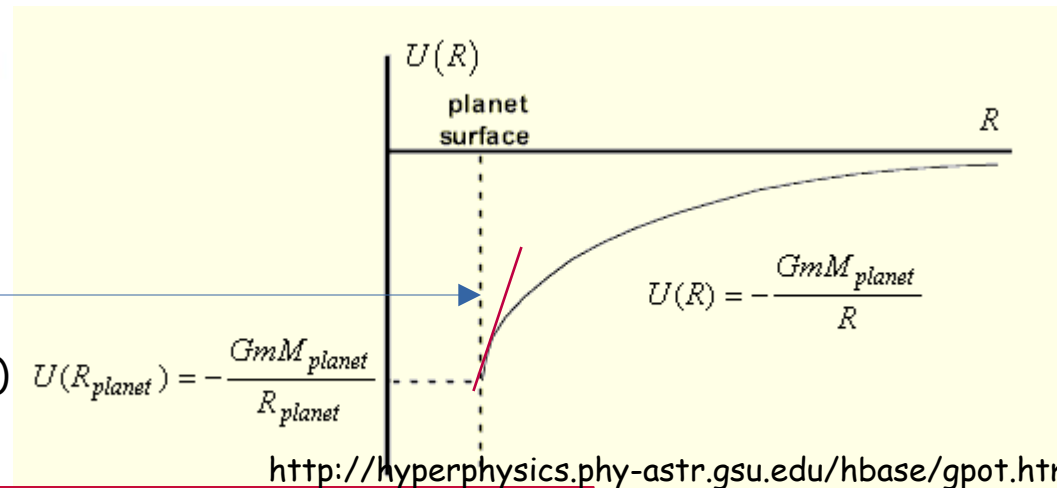
and the law of gravitation $F = -\frac{GMm}{r^2} \hat{r}$

with the choice of the zero of potential energy at infinite distance where the force approaches zero, the gravitational potential energy is the work done to bring an object from infinity to radius r :

$$U(r) = -\int_{\infty}^r -\frac{GMm}{r'^2} dr' = -\frac{GMm}{r}$$

Form of integral

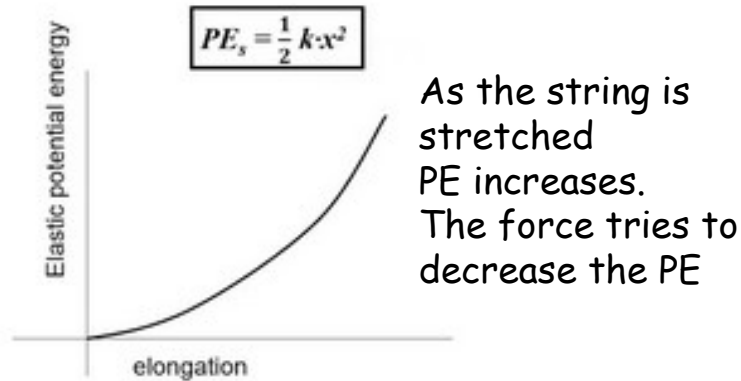
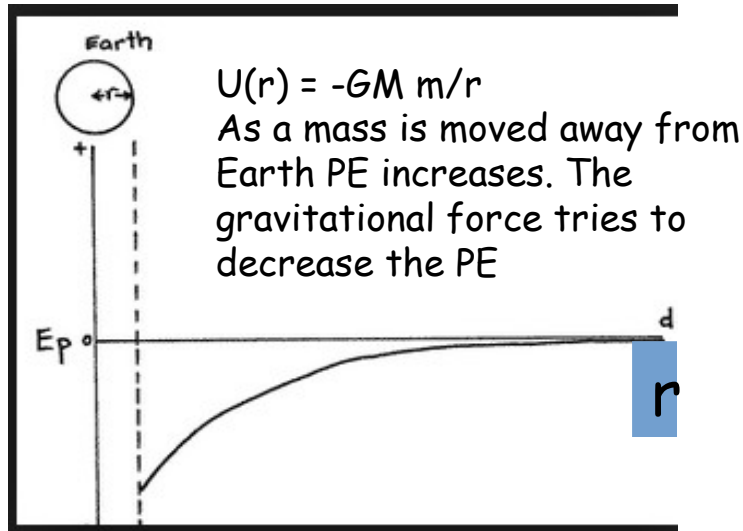
$\hat{r}$ represents a unit vector in the outward radial direction



<http://hyperphysics.phy-astr.gsu.edu/hbase/gpot.html>



→ Graphs PE as a function of distance



→ Take the negative of $U = PE$ as a function of distance. What do you get? This is true for any PE and restoring force

$f_{pr} = -dU/dr$

The restoring forces derive from a $PE = U$

Show on the graphs F is the slope of U and point to Lower PE



1. Gravitational

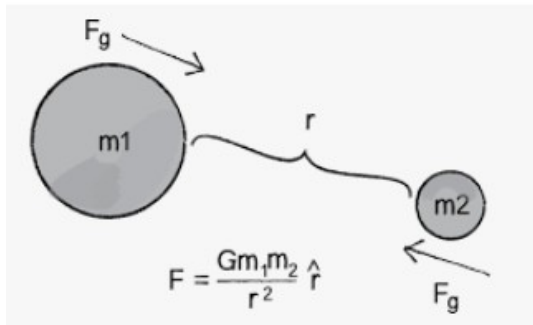
$$G = 6.7 \times 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

$$U_g = \frac{-GMm}{r}$$

M = mass of object 1

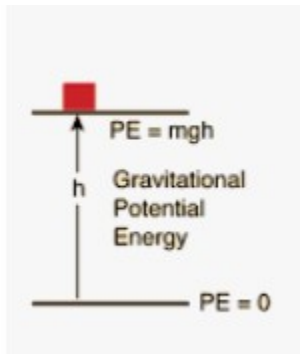
m = mass of object 2

r = center to center distance between M and m



The potential energy is 0 at infinity
The restoring force is $F = G m_1 m_2 / r^2$
When r is increased the PE increases

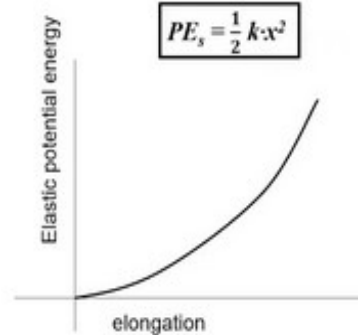
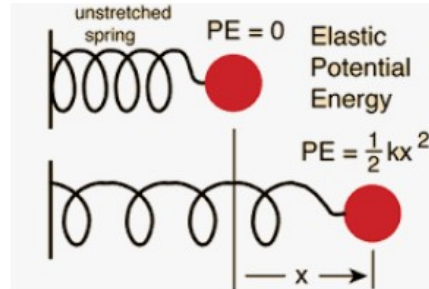
If close to Earth



The restoring force is $F = -mg$
 $PE = 0$ when $h = 0$
When h increases, PE increases

2. Elastic

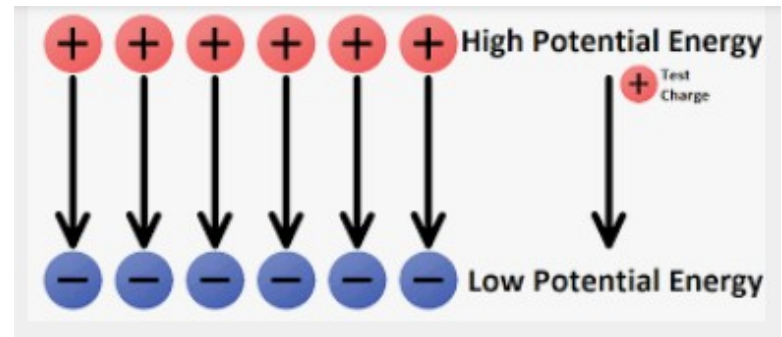
Spring



As the string is stretched PE increases. The force tries to decrease the PE

The potential energy is 0 at $x = 0$
The restoring force is $F = -Kx$
When x is increased the PE increases

Electric PE

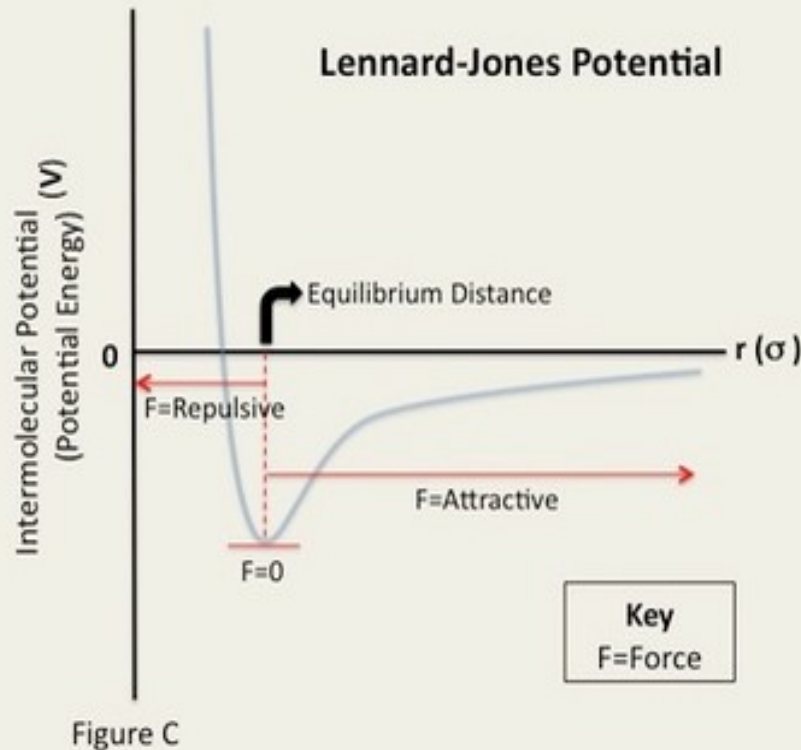


The potential energy is 0 at $h = 0$
The restoring force is $F = -qE$ and $PE = qEh$
When x is increased the PE increases



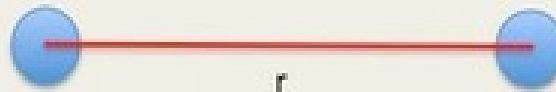
PE at a molecular level (see your chem class)

the Lennard-Jones potential is often used to describe the properties of gases and simple fluids and to model dispersive and repulsive interactions in molecular models. It is especially accurate for noble gas atoms and methane





1. Both balls are an infinite distance apart and are not interacting



2. The balls are brought closer together with minimal energy input to a certain distance, r . At this distance, the balls have an attractive force between them



3. The attractive force between the two objects brings the objects even further together until they reach an equilibrium distance apart at which their minimum bonding potential is reached.



4. To further decrease the distance between both objects, additional energy is required because as the balls overlap, repulsive forces act and push both balls further apart. At these distances, the force of repulsion is greater than the force of attraction.

Figure A



Power

Work does not depend on time interval

The rate at which energy is transferred is important in the design and use of practical device

The time rate of energy transfer is called power

The average power is given by

$$\overline{P} = \frac{W}{\Delta t}$$

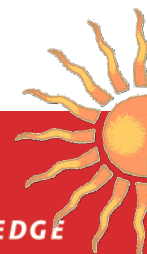
- when the method of energy transfer is work



Table 6.4 Human Metabolic Rates^a

Activity	Rate (watts)
Running (15 km/h)	1340 W
Skiing	1050 W
Biking	530 W
Walking (5 km/h)	280 W
Sleeping	77 W

^aFor a young 70-kg male.



Units of Power

The SI unit of power is called the watt

- $1 \text{ watt} = 1 \text{ joule} / \text{second} = 1 \text{ kg} \cdot \text{m}^2 / \text{s}^3$

A unit of power in the US Customary system is horsepower

- $1 \text{ hp} = 550 \text{ ft} \cdot \text{lb/s} = 746 \text{ W}$

Units of power can also be used to express units of work or energy

- $1 \text{ kWh} = (1000 \text{ W})(3600 \text{ s}) = 3.6 \times 10^6 \text{ J}$



for power examples

1 watt for flashlight battery

100 watts for a human or incandescent light

1 hp for a typical horse or athlete exercising for short period of time

1 kilowatt for a small house = **energy from the Sun for a square 1 m x 1m**

100hp for a small car

1 megawatt for electric power for small town (1000 houses)

1 gigawatt for large power plant (1000,000)

2 terawatts for the world (4/10 for the US)

Source: Richard A. Muller, Physics for future presidents.



Kwh is a unit of energy :

Multiply the power (in kW) by the time (in hours)

Example: a microwave use 800 watts. If you run it for 2 hours and if the electric company charges 10 cents/kwh. How much you are going to pay ?

Now: Miami → 15 cents
NYC → 30 cents
Hawaii → 40 cents

Electricity is cheap !



1 Kwh is a unit for energy = $1000 \times 3600 \text{ J}$
or about 4 million J

1 good calorie or 1 Cal = 4000J so 1 Cal is 1Wh about
Or 1 KWh is about 1000 Cal.

Don't confuse the chem cal and the food Cal. 1 Cal = 1000 cal

We are like a 100W light bulb.

In one day we use about _____ Wh
Which is about _____ Cal (1Wh is 1 Cal).

Lots of energy in food! 1 glass of milk = 150Cal = 1 soda can.
It takes 30 min hard gym to burn 150 Cal.



Can you lose weight doing exercises ? Or is it better to cut on Food ? What's your guess ?

1 can of soda is about 150 Calories
(so is 1 glass of orange juice or milk)

You can burn this much of calories
In half an hour of cardio exercise



In your body you have 7 Cal per gram.
So if you cut your Calories from 2,000 to 1,500 calories per day.
How many grams do you lose per day ?
Per week ? How many pounds per week ? (500g = 1 pound)

If you exercise for 1 hour how many calories do you lose ?
How many grams is that ? (compare with previous)
So in a week you lose ? So?



Other unit for power is horsepower or hp

Introduced by James Watt.

A horse can lift 330 pounds weight vertically

For a distance of 100 ft in one minute

Or $P = 330 \times 100$ foot-pounds per minute

Or $P = 33,000$ foot-pounds per minute

We can produce the same power but for a short time.

Usually, we can produce $4/7$ hp with sustained effort.

(like heavy duty exercise not yoga)

$1/7$ is used to do work the other part is lost to heat.

($1/7$ is 25% of $4/7$, so we are only 25% efficient)

1Hp is 746 W about 1kW

See p. 20 table

So 1 horse can lift _____ pounds over 1 foot



So we can produce $4/7$ hp during a sustained exercise.



BUT NOT FOR VERY LONG !!
Unlike a horse.

Energy and Power Units

Energy units

joules (J)

calories (cal, kcal)

kilowatt-hours (kWh)

British thermal
units (Btu)

electron-volts (eV)

Barrels of oil
equivalent (BOE)

Power units

watts (W; $1 \text{ W} = 1 \text{ J/s}$)

horsepower (hp; $1 \text{ hp} = 746 \text{ W}$)

Million barrels of
oil equivalent per
day (MBOE/day)

Humankind's energy
consumption rate:
 16 TW ($16 \times 10^{12} \text{ W}$)



$$\text{Average Power} = \frac{\text{Total Work Done}}{\text{Total Time}}$$

This definition holds true regardless of whether the force is constant or variable. It's a more general and accurate way to express average power in all scenarios.

$$\bar{P} = \frac{W}{\Delta t} = \frac{F\Delta x}{\Delta t} = F \bar{v}$$

→ you can use this equation if the Force is constant and if there is A constant acceleration. (0 or constant)

The force is // to displacement

If the force is not constant. You need to integrate to find average power.

$$P_{avg} = \frac{1}{T} \int_0^T \vec{F}(t) \cdot \vec{v}(t) dt$$

Where:

- $\vec{F}(t)$ is the force vector as a function of time,
- $\vec{v}(t)$ is the velocity vector as a function of time.

$$\vec{F} \cdot \vec{v} = Fv \cos \theta$$

Is the instantaneous power

$$P = \lim_{\Delta t \rightarrow 0} \frac{W}{\Delta t} = \frac{dW}{dt} = \vec{F} \cdot \frac{d\vec{r}}{dt} = \vec{F} \cdot \vec{v}$$



Go back to this problem after next unit.

Net work = change in KE

Example

Example 3.10

In Examples 2.2 and 3.5, we computed the acceleration, force and work for a 1,000-kg car that goes from 0 to 27 m/s in 10 s. We can now determine the required power output of the engine. What is its value?

Way 1: Easier average power = net work/time

Way 2: average power = force $\times$ average speed

Average speed = $(V_1 + V_2)/2$ if constant acceleration

And force = ma



Calculating Power: Cycling

What's the minimum power needed to bicycle up a 10% (6°) slope at 9 mph (4 meters/second)?

Mass of bicycle and rider 80 kg

Use average power =
Force $\times$ speed

And force is the force you
Provide against gravity
No acceleration. $mg\sin(\text{angle})$



Calculating Power: Driving

A gallon of gasoline contains about 40 kWh of energy. At what rate is your car consuming fuel energy if you're driving at 60 mph and getting 30 miles per gallon?



Calculating Power: Driving

A gallon of gasoline contains about 40 kWh of energy. At what rate is your car consuming fuel energy if you're driving at 60 mph and getting 30 miles per gallon?

30 miles $\rightarrow$ 1 gallon
60 miles $\rightarrow$ 2 gallons

2 gallons contain 80 kWh
Or 80,000 kWh $\rightarrow$ 1 hour
Rate = 80,000 kWh/h = 80,000 Watts
About 100 hp

The average horsepower for a car is
About 200 hp

483 to 542 hp
2024 Maserati GranTurismo

657 hp
2023 Lamborghini Urus

819 hp
2023 Ferrari 812 Competizione



About 1000 hp for F1
120 hp for my venue :-)

65 to 80 horsepower for
hybrid

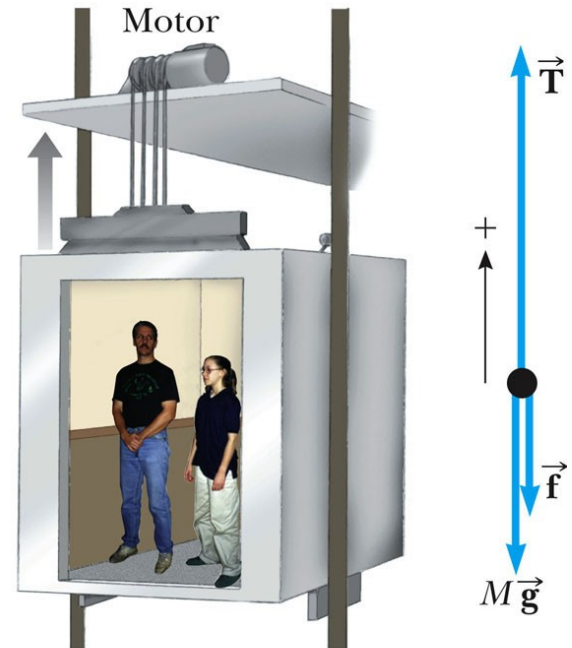


Power Delivered by an Elevator Motor

□ A 1000-kg elevator carries a maximum load of 800 kg. A constant frictional force of 4000 N retards its motion upward. What minimum power must the motor deliver to lift the fully loaded elevator at a constant speed of 3 m/s?

Hint: you are asked to find the Power delivered by T. Find T
First: up = down since $a = 0$

Then use power = force x speed



Power Delivered by an Elevator Motor

□ A 1000-kg elevator carries a maximum load of 800 kg. A constant frictional force of 4000 N retards its motion upward. What minimum power must the motor deliver to lift the fully loaded elevator at a constant speed of 3 m/s?

$$F_{net,y} = ma_y$$

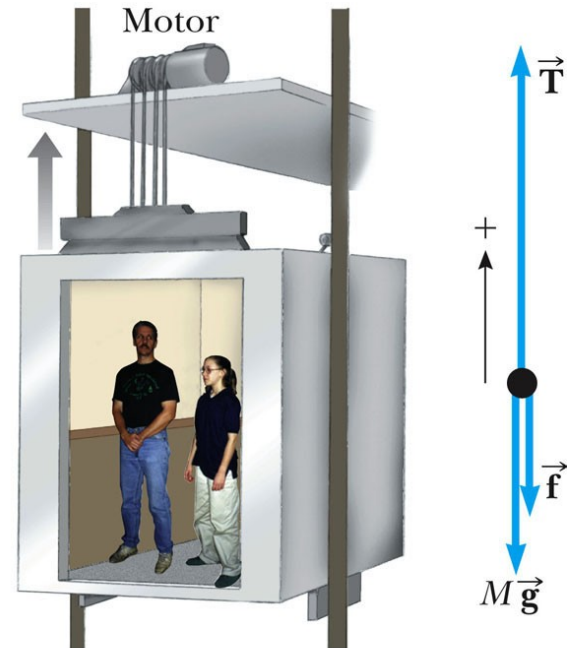
$$T - f - Mg = 0$$

$$T = f + Mg = 2.16 \times 10^4 \text{ N}$$

$$P = Fv = (2.16 \times 10^4 \text{ N})(3 \text{ m/s})$$

$$6.48 \times 10^4 \text{ W}$$

$$P = 64.8 \text{ kW} = 86.9 \text{ hp}$$



(b)



Power and Energy

- **Power** is the *rate* at which work is done or at which energy is used or produced. If work ΔW is done in time Δt , then the **average power** over this time is

$$\bar{P} = \frac{\Delta W}{\Delta t} \quad (\text{average power})$$

- When the rate changes continuously, the **instantaneous power** is

$$P = \lim_{\Delta t \rightarrow 0} \frac{\Delta W}{\Delta t} = \frac{dW}{dt}$$

- Power is measured in **watts** (W), with $1 \text{ W} = 1 \text{ J/s}$.
- Total work or energy follows from power by multiplying (for constant power) or integrating (for varying power):

$$W = P \Delta t \quad \text{or} \quad W = \int_{t_1}^{t_2} P dt$$



Huge Idea in Physics! Conservation of Energy

Work-energy theorem: $\Delta K = W_{\text{net}}$

Potential energy: $\Delta U = -W_{\text{conservative}} \Rightarrow W_{\text{conservative}} = -\Delta U$

If only conservative forces act: $\Delta K = -\Delta U$

So: $\Delta K + \Delta U = 0$

Conservation of Mechanical Energy:

When only conservative forces act, the total mechanical energy $K + U$ does not change

